MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

IOC CONTROL MEMORY MODULE, KEY CODE AWF

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic Input/Output Controller (IOC) control memory module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 4.227 failures per million operating hours for class I or 10.881 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be AWF.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
B1-B6	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Voc stress	IV	S	-------	The module shall pass al			remaining tests		-------
D1	Ibd	V	S	0.0	4.0	0.0	4.0	0.0	4.0	mA
D2	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D3	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D4,D5	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D6	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D7	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D8	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D9	Ibd	V	S	0.0	1.1	0.0	1.1	0.0	1.1	mA
D10	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D11	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D12	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D13,D14	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D15,D16	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D17-D19	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D20-D23	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D24	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D25	Ibd	V	S	0.0	1.2	0.0	1.2	0.0	1.2	mA
D26	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D27	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D28,D29	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D30	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	mA
D31	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D32	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
E1	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
E2	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E3	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E4	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E5	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E6, E7	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E8	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E9, E10	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E11	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E12, E13	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E14	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E15	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E16	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E17, E18	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E19	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E20, E21	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E22	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E23, E24	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E25-E33	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E34	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E35	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E36, E37	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E38	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E39, E40	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E41	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E42, E43	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E44	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E45, E46	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E47	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E48, E49	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E50	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E51, E52	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 8 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
E53	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E54,E55	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E56	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E57	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E58-E63	Ios	VI	S	-325.0	-100.0	-325.0	-100.0	-325.0	-100.0	mA
E64	Ios	VI	S	-100.0	-20.0	-100.0	-20.0	-100.0	-20.0	mA
E65-E72	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E73-E80	Ios	VI	S	-325.0	-100.0	-325.0	-100.0	-325.0	-100.0	mA
F1	Iih	VII	X	-0.0	0.2	-0.0	0.2	-0.0	0.2	mA
F2	Iih	VII	X	-0.0	0.04	-0.0	0.04	-0.0	0.04	mA
F3	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F4,F5	Iih	VII	X	-0.4	0.04	-0.4	0.04	-0.4	0.04	mA
F6	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F7	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F8	Iih	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F9	Iih	VII	X	-0.45	0.05	-0.45	0.05	-0.45	0.05	mA
F10	Iih	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F11	Iih	VII	X	-0.4	0.04	-0.4	0.04	-0.4	0.04	mA
F12	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F13,F14	Iih	VII	X	-0.4	0.04	-0.4	0.04	-0.4	0.04	mA
F15,F16	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F17-F24	Iih	VII	X	-0.46	0.1	-0.46	0.1	-0.46	0.1	mA
F25-F27	Iih	VII	X	-0.4	0.04	-0.4	0.04	-0.4	0.04	mA
F28-F31	Iih	VII	X	-0.6	0.06	-0.6	0.06	-0.6	0.06	mA
F32	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F33	Iih	VII	X	-2.0	0.09	-2.0	0.09	-2.0	0.09	mA
F34	Iih	VII	X	-1.62	0.04	-1.62	0.04	-1.62	0.04	mA
F35,F36	Iih	VII	X	-1.87	0.07	-1.87	0.07	-1.87	0.07	mA
F37	Iih	VII	X	-0.4	0.04	-0.4	0.04	-0.4	0.04	mA
F38-F40	Iih	VII	X	-1.62	0.04	-1.62	0.04	-1.62	0.04	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F41	Ilh	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F42-F49	Ilh	VII	X	-0.26	0.00	-0.26	0.00	-0.26	0.00	mA
F50	Ilh	VII	X	-2.0	0.04	-2.0	0.04	-2.0	0.04	mA
F51-F56	Ilh	VII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
F57-F62	Ilh	VII	X	-3.47	0.09	-3.47	0.09	-3.47	0.09	mA
F63,F64	Ilh	VII	X	-1.87	0.07	-1.87	0.07	-1.87	0.07	mA
F65	Ilh	VII	X	-2.0	0.04	-2.0	0.04	-2.0	0.04	mA
F66,F67	Ilh	VII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
F68	Ilh	VII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	mA
F69	Ilh	VII	X	-1.9	0.06	-1.9	0.06	-1.9	0.06	mA
F70	Ilh	VII	X	-2.0	0.04	-2.0	0.04	-2.0	0.04	mA
F71-F78	Ilh	VII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
G1	Ill	VIII	X	-0.0	0.2	-0.0	0.2	-0.0	0.2	mA
G2	Ill	VIII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
G3	Ill	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G4,G5	Ill	VIII	X	-0.4	0.04	-0.4	0.04	-0.4	0.04	mA
G6	Ill	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G7	Ill	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G8	Ill	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G9	Ill	VIII	X	-0.45	0.05	-0.45	0.05	-0.45	0.05	mA
G10	Ill	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G11	Ill	VIII	X	-0.4	0.04	-0.4	0.04	-0.4	0.04	mA
G12	Ill	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G13,G14	Ill	VIII	X	-0.4	0.04	-0.4	0.04	-0.4	0.04	mA
G15,G16	Ill	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G17-G24	Ill	VIII	X	-0.46	0.1	-0.46	0.1	-0.46	0.1	mA
G25-G27	Ill	VIII	X	-0.4	0.04	-0.4	0.04	-0.4	0.04	mA
G28-G31	Ill	VIII	X	-0.6	0.06	-0.6	0.06	-0.6	0.06	mA
G32	Ill	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G33	Ill	VIII	X	-2.8	0.09	-2.8	0.09	-2.8	0.09	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
G34	Iil	VIII	X	-1.62	0.04	-1.62	0.04	-1.62	0.04	mA
G35,G38	Iil	VIII	X	-1.87	0.07	-1.87	0.07	-1.87	0.07	mA
G37	Iil	VIII	X	-0.4	0.04	-0.4	0.04	-0.4	0.04	mA
G38-G40	Iil	VIII	X	-1.82	0.04	-1.82	0.04	-1.82	0.04	mA
G41	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G42-G49	Iil	VIII	X	-0.26	0.08	-0.26	0.08	-0.26	0.08	mA
G50	Iil	VIII	X	-2.0	0.04	-2.0	0.04	-2.0	0.04	mA
G51-G56	Iil	VIII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
G57-G62	Iil	VIII	X	-3.47	0.09	-3.47	0.09	-3.47	0.09	mA
G63,G64	Iil	VIII	X	-1.87	0.07	-1.87	0.07	-1.87	0.07	mA
G65	Iil	VIII	X	-2.0	0.04	-2.0	0.04	-2.0	0.04	mA
G66,G67	Iil	VIII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
G68	Iil	VIII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	mA
G69	Iil	VIII	X	-1.0	0.06	-1.0	0.06	-1.0	0.06	mA
G70	Iil	VIII	X	-2.0	0.04	-2.0	0.04	-2.0	0.04	mA
G71-G78	Iil	VIII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
H1-H88	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J89	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K1,K2	Iozh	XI	X	-0.02	0.02	-0.02	0.02	-0.02	0.02	mA
K3-K10	Iozh	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
K11,K12	Iozl	XI	X	-0.02	0.02	-0.02	0.02	-0.02	0.02	mA
K13-K20	Iozl	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
L1-L3	Vlc	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L4,L5	Vlc	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L6,L7	Vlc	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L8-L11	Vlc	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L12	Vlc	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ∓ 5°C		Class II 4/ -55 ± 5°C 85 ∓ 5°C		Units
				Min	Max	Min	Max	Min	Max	
L13-L24	Vlc	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L25	Vlc	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L26	Vlc	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L27-L32	Vlc	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
M1	PSFC	XIII	S	5.9	11.0	5.9	11.0	5.9	11.0	uF
M2	PSFC	XIII	S	6.4	12.0	6.4	12.0	6.4	12.0	uF
N	Ioo	XIV	X	640	1637	640	1637	640	1637	mA
P1,P2	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	S	See table XVI						
Q3,Q4	Dynamic	XVI	Q	See table XVI						
R1	C1/o	XVIII	Q	8	125	8	125	8	125	pF
R2	C1/o	XVIII	Q	4	125	4	125	4	125	pF
R3	C1/o	XVIII	Q	2	100	2	100	2	100	pF
R4,R5	C1/o	XVIII	Q	4	75	4	75	4	75	pF
R6-R8	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R9	C1/o	XVIII	Q	2	100	2	100	2	100	pF
R10	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R11	C1/o	XVIII	Q	4	100	4	100	4	100	pF
R12	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R13,R14	C1/o	XVIII	Q	4	125	4	125	4	125	pF
R15,R16	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R17	C1/o	XVIII	Q	4	100	4	100	4	100	pF
R18	C1/o	XVIII	Q	4	125	4	125	4	125	pF
R19	C1/o	XVIII	Q	4	100	4	100	4	100	pF
R20-R23	C1/o	XVIII	Q	6	100	6	100	6	100	pF
R24	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R25	C1/o	XVIII	Q	6	100	6	100	6	100	pF
R26	C1/o	XVIII	Q	4	100	4	100	4	100	pF

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R27	CI/O	XVIII	Q	2	125	2	125	2	125	pF
R28-R30	CI/O	XVIII	Q	4	75	4	75	4	75	pF
R31	CI/O	XVIII	Q	6	100	6	100	6	100	pF
R32	CI/O	XVIII	Q	4	75	4	75	4	75	pF
R33	CI/O	XVIII	Q	10	125	10	125	10	125	pF
R34,R35	CI/O	XVIII	Q	10	150	10	150	10	150	pF
R36,R37	CI/O	XVIII	Q	10	125	10	125	10	125	pF
R38,R39	CI/O	XVIII	Q	10	150	10	150	10	150	pF
R40	CI/O	XVIII	Q	10	75	10	75	10	75	pF
R41	CI/O	XVIII	Q	8	100	8	100	8	100	pF
R42,R43	CI/O	XVIII	Q	8	125	8	125	8	125	pF
R44	CI/O	XVIII	Q	8	100	8	100	8	100	pF
R45-R48	CI/O	XVIII	Q	8	125	8	125	8	125	pF
R49	CI/O	XVIII	Q	4	75	4	75	4	75	pF
R50	CI/O	XVIII	Q	4	100	4	100	4	100	pF
R51-R54	CI/O	XVIII	Q	4	75	4	75	4	75	pF
R55	CI/O	XVIII	Q	4	125	4	125	4	125	pF
R56	CI/O	XVIII	Q	4	150	4	150	4	150	pF
R57,R58	CI/O	XVIII	Q	4	100	4	100	4	100	pF
R59,R60	CI/O	XVIII	Q	4	75	4	75	4	75	pF
R61,R62	CI/O	XVIII	Q	2	75	2	75	2	75	pF
T1,T2	PSTR	XIX	Q	0	400	0	400	0	400	mV

- See footnotes at end of table.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/504-1 for class I and M28787/504-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

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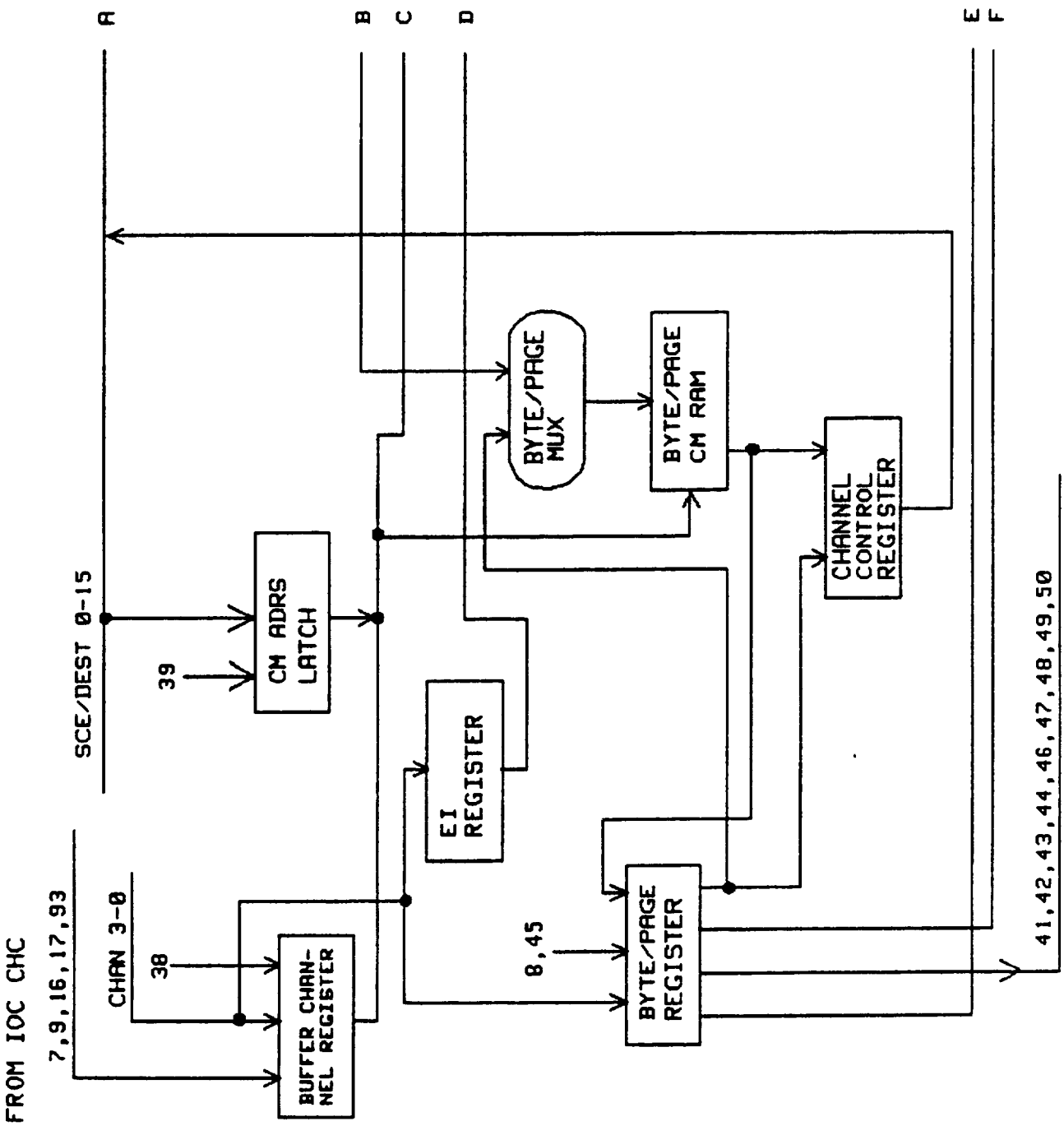


FIGURE 1. AWF functional block diagram.

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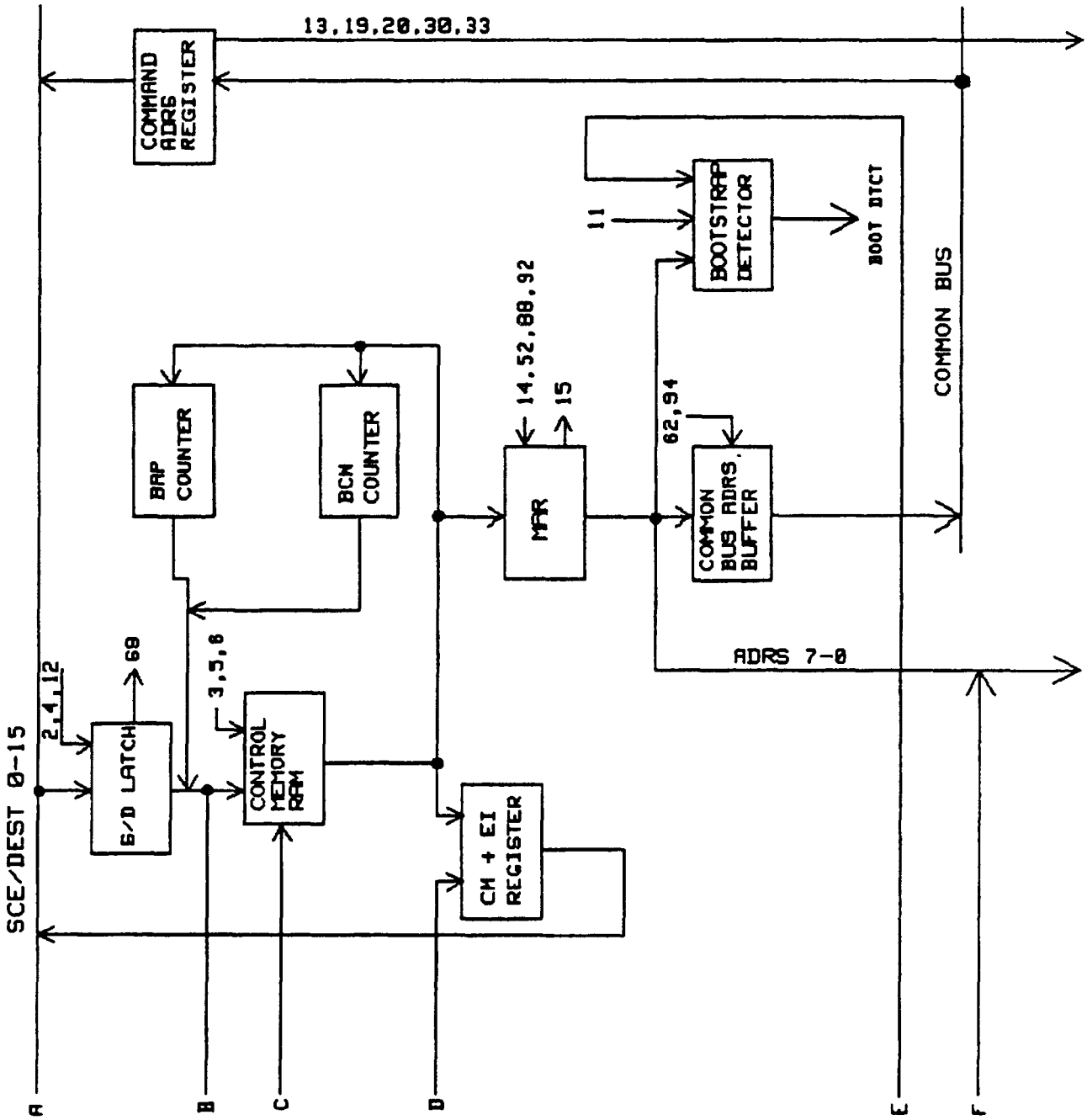


FIGURE 1. AWF functional block diagram - Continued.

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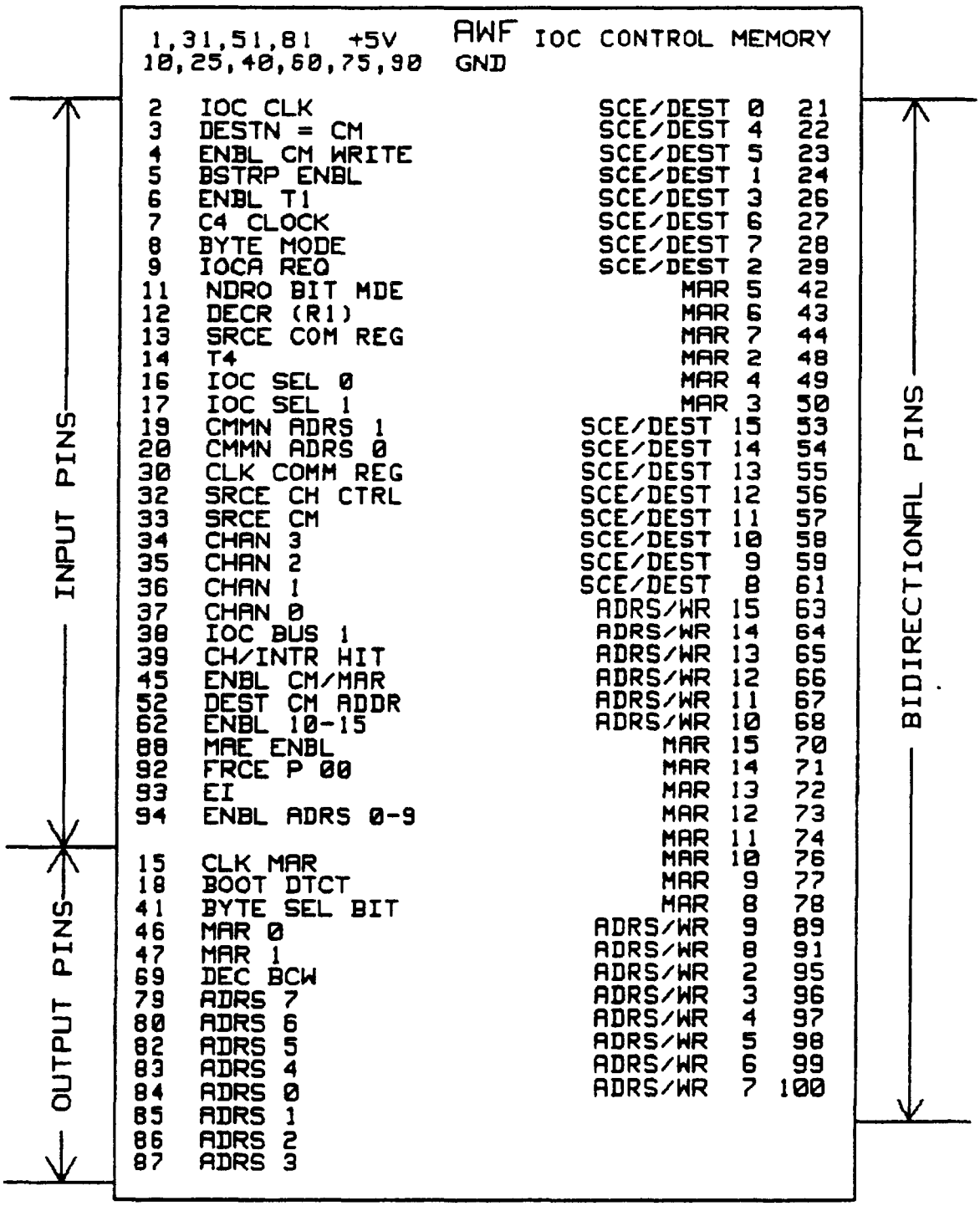


FIGURE 2. AWF pin assignment.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M

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through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

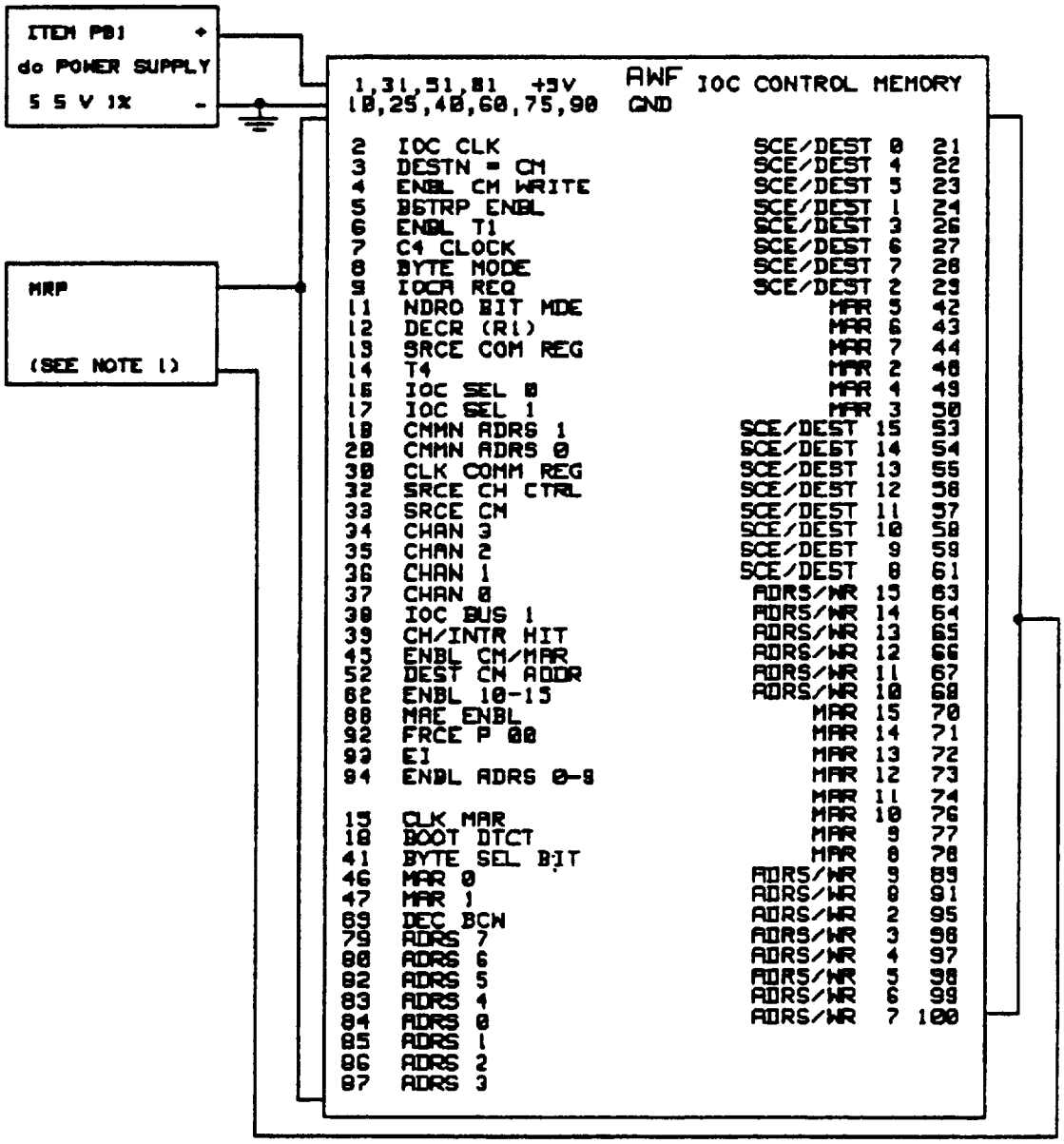
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. AWF operating temperature and component temperature test circuit.

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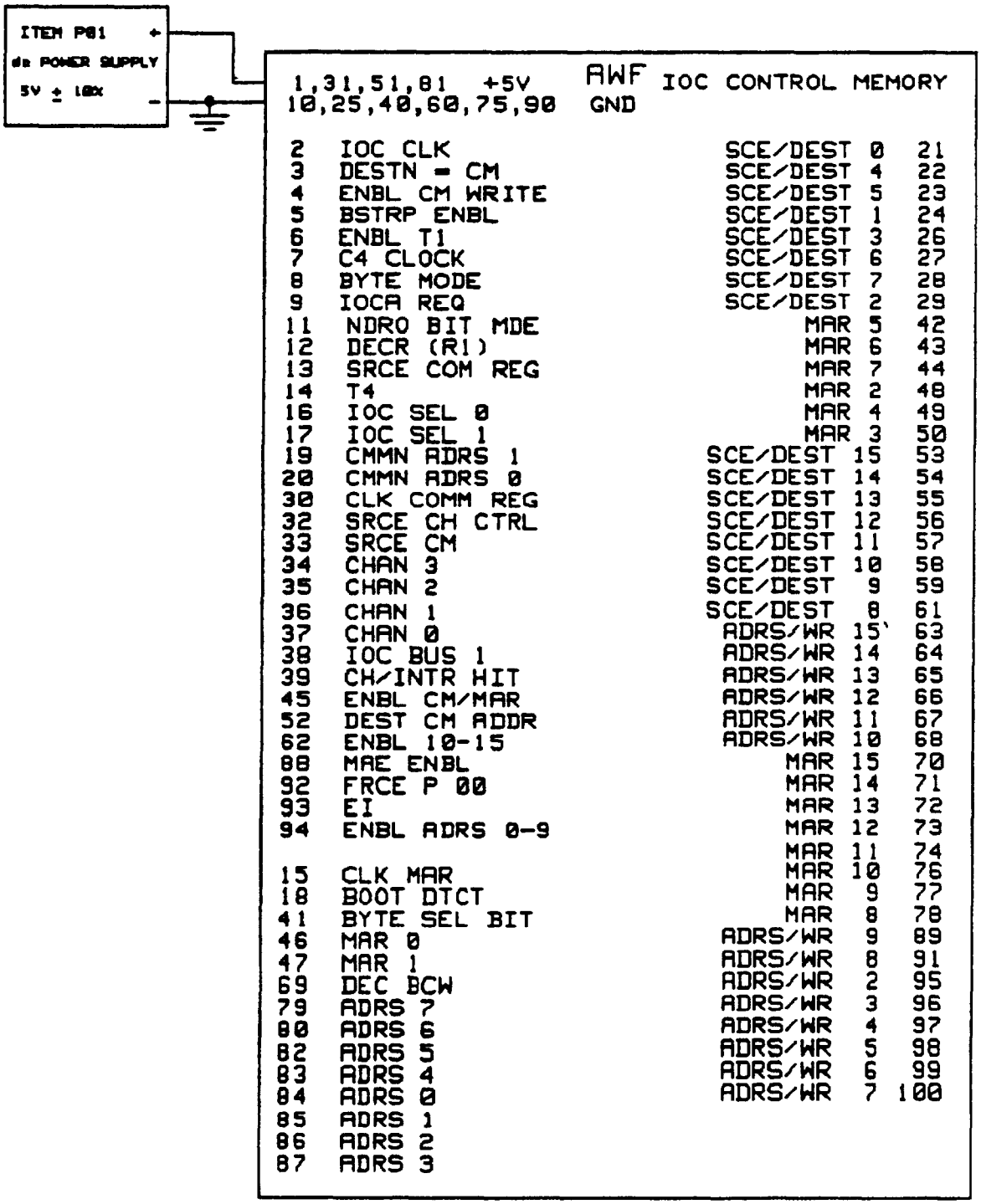
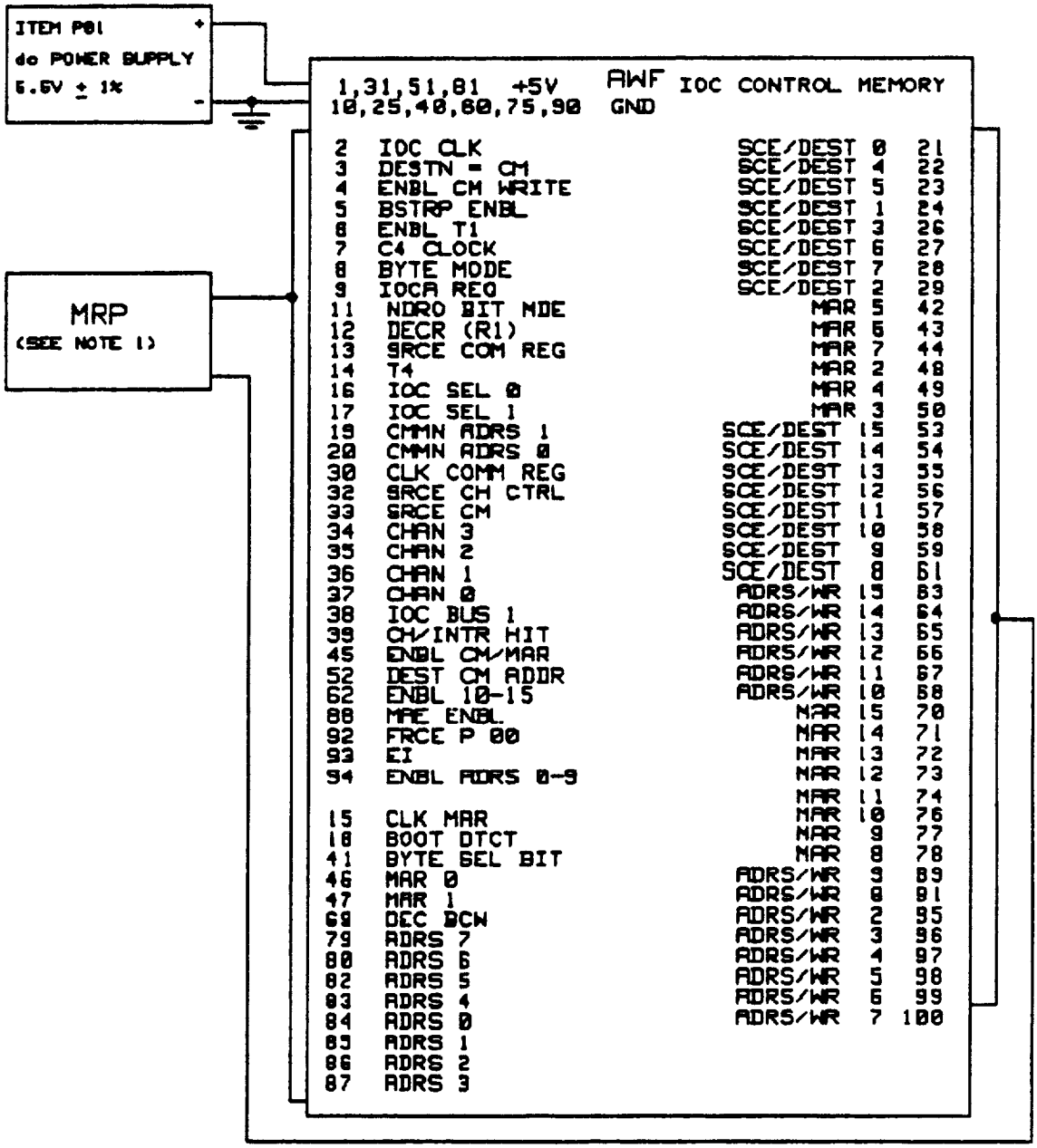


FIGURE 4. AWF vibration test circuit.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. AWF life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 9.00 W (max); 5.94 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
IOC CLK	-1.2	0.4	2.4	-8.0	0.2	4.0	125
DESTN=CM	-1.2	0.4	2.4	-0.8	0.04	0.2	125
ENBL CM WR	-1.2	0.4	2.4	-0.6	0.02	0.1	100
BSTRP ENBL	-1.5	0.4	2.4	-0.4	0.04	0.2	75
ENBL T1	-1.5	0.4	2.4	-0.4	0.04	0.2	75
C4 CLK	-1.2	0.4	2.4	-0.6	0.02	0.1	75
BYTE MODE	-1.2	0.4	2.4	-2.0	0.05	1.0	75
IOCA REQ	-1.5	0.4	2.4	-0.4	0.02	0.1	75
NDRO BIT MDE	-1.5	0.4	2.4	-0.45	0.05	1.1	100
DECR (R1)	-1.5	0.4	2.4	-0.4	0.02	0.1	75
SRCE COM REG	-1.5	0.4	2.4	-0.4	0.04	0.2	100
T4	-1.2	0.4	2.4	-2.0	0.05	1.0	75
IOC SEL 1	-1.5	0.4	2.4	-0.4	0.04	0.2	125
IOC SEL 0	-1.5	0.4	2.4	-0.4	0.04	0.2	125

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
CMMN ADRS 1	-1.5	0.4	2.4	-0.2	0.02	0.1	75
CMMN ADRS 0	-1.5	0.4	2.4	-0.2	0.02	0.1	75
CLK COMM REG	-1.5	0.4	2.4	-0.4	0.04	0.2	100
SRCE CH CTRL	-1.5	0.4	2.4	-0.4	0.04	0.2	125
SRCE CM	-1.5	0.4	2.4	-0.4	0.04	0.2	100
CHAN 3	-1.5	0.4	2.4	-0.6	0.06	0.3	100
CHAN 2	-1.5	0.4	2.4	-0.6	0.06	0.3	100
CHAN 1	-1.5	0.4	2.4	-0.6	0.06	0.3	100
CHAN 0	-1.5	0.4	2.4	-0.6	0.06	0.3	100
IOC BUS 1	-1.5	0.4	2.4	-0.2	0.02	0.1	75
CH/INTR HIT	-1.2	0.4	2.4	-2.8	0.09	1.2	100
ENBL CM/MAR	-1.5	0.4	2.4	-0.4	0.04	0.2	100
DEST CM ADDR	-1.2	0.4	2.4	-0.6	0.02	0.1	125
ENBL 10-15	-1.2	0.4	2.4	-2.0	0.04	0.2	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MAE ENBL	-1.2	0.4	2.4	-2.0	0.04	0.2	75
FRCE P 00	-1.2	0.4	2.4	-4.0	0.1	2.0	75
EI	-1.2	0.4	2.4	-1.8	0.06	0.3	100
ENBL ADR 0-9	-1.2	0.4	2.4	-2.0	0.04	0.2	75
SCE/DEST 0	NA	0.4	2.4	-0.46	0.1	NA	125
SCE/DEST 4	NA	0.4	2.4	-0.46	0.1	NA	150
SCE/DEST 5	NA	0.4	2.4	-0.46	0.1	NA	150
SCE/DEST 1	NA	0.4	2.4	-0.46	0.1	NA	125
SCE/DEST 3	NA	0.4	2.4	-0.46	0.1	NA	125
SCE/DEST 6	NA	0.4	2.4	-0.46	0.1	NA	150
SCE/DEST 7	NA	0.4	2.4	-0.46	0.1	NA	150
SCE/DEST 2	NA	0.4	2.4	-0.46	0.1	NA	75
MAR 5	NA	0.4	2.4	-1.62	0.04	NA	NA
MAR 6	NA	0.4	2.4	-1.87	0.07	NA	NA

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MAR 7	NA	0.4	2.4	-1.87	0.07	NA	NA
MAR 2	NA	0.4	2.4	-1.62	0.04	NA	NA
MAR 4	NA	0.4	2.4	-1.62	0.04	NA	NA
MAR 3	NA	0.4	2.4	-1.62	0.04	NA	NA
SCE DEST 15	NA	0.4	2.4	-0.26	0.08	NA	100
SCE/DEST 14	NA	0.4	2.4	-0.26	0.08	NA	125
SCE/DEST 13	NA	0.4	2.4	-0.26	0.08	NA	125
SCE/DEST 12	NA	0.4	2.4	-0.26	0.08	NA	100
SCE/DEST 11	NA	0.4	2.4	-0.26	0.08	NA	125
SCE/DEST 10	NA	0.4	2.4	-0.26	0.08	NA	125
SCE/DEST 9	NA	0.4	2.4	-0.26	0.08	NA	125
SCE/DEST 8	NA	0.4	2.4	-0.26	0.08	NA	125
ADRS/WR 15	NA	0.4	2.4	-0.25	0.07	NA	75
ADRS/WR 14	NA	0.4	2.4	-0.25	0.07	NA	100

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
ADRS/WR 13	NA	0.4	2.4	-0.25	0.07	NA	75
ADRS/WR 12	NA	0.4	2.4	-0.25	0.07	NA	75
ADRS/WR 11	NA	0.4	2.4	-0.25	0.07	NA	75
ADRS/WR 10	NA	0.4	2.4	-0.25	0.07	NA	75
MAR 15	NA	0.4	2.4	-3.47	0.09	NA	NA
MAR 14	NA	0.4	2.4	-3.47	0.09	NA	NA
MAR 13	NA	0.4	2.4	-3.47	0.09	NA	NA
MAR 12	NA	0.4	2.4	-3.47	0.09	NA	NA
MAR 11	NA	0.4	2.4	-3.47	0.09	NA	NA
MAR 10	NA	0.4	2.4	-3.47	0.09	NA	NA
MAR 9	NA	0.4	2.4	-1.87	0.07	NA	NA
MAR 8	NA	0.4	2.4	-1.87	0.07	NA	NA
ADRS/WR 9	NA	0.4	2.4	-0.25	0.07	NA	125
ADRS/WR 8	NA	0.4	2.4	-0.25	0.07	NA	150

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
ADRS/WR 2	NA	0.4	2.4	-0.25	0.07	NA	100
ADRS/WR 3	NA	0.4	2.4	-0.25	0.07	NA	100
ADRS/WR 4	NA	0.4	2.4	-0.25	0.07	NA	75
ADRS/WR 5	NA	0.4	2.4	-0.25	0.07	NA	75
ADRS/WR 6	NA	0.4	2.4	-0.25	0.07	NA	75
ADRS/WR 7	NA	0.4	2.4	-0.25	0.07	NA	75

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
CLK MAR	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
BOOT DTCT	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
BYTE SEL BIT	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
MAR 0	2.4	0.5	-0.16	1.6	0.02	0.02	-30.0	NA
MAR 1	2.4	0.5	-0.16	1.6	0.02	0.02	-30.0	NA
DEC BCW	2.4	0.5	-0.16	1.6	NA	NA	-20.0	NA
ADRS 7	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA
ADRS 6	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA
ADRS 5	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA
ADRS 4	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA
ADRS 0	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA
ADRS 1	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA
ADRS 2	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA
ADRS 3	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
SCE/DEST 0	2.4	0.5	-0.16	1.6	NA	NA	-30.0	125
SCE/DEST 4	2.4	0.5	-0.16	1.6	NA	NA	-30.0	150
SCE/DEST 5	2.4	0.5	-0.16	1.6	NA	NA	-30.0	150
SCE/DEST 1	2.4	0.5	-0.16	1.6	NA	NA	-30.0	125
SCE/DEST 3	2.4	0.5	-0.16	1.6	NA	NA	-30.0	125
SCE/DEST 6	2.4	0.5	-0.16	1.6	NA	NA	-30.0	150
SCE/DEST 7	2.4	0.5	-0.16	1.6	NA	NA	-30.0	150
SCE/DEST 2	2.4	0.5	-0.16	1.6	NA	NA	-30.0	125
MAR 5	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
MAR 6	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
MAR 7	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
MAR 2	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
MAR 4	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
MAR 3	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
SCE/DEST 15	2.4	0.5	-0.16	1.6	NA	NA	-30.0	100
SCE/DEST 14	2.4	0.5	-0.16	1.6	NA	NA	-30.0	125
SCE/DEST 13	2.4	0.5	-0.16	1.6	NA	NA	-30.0	125
SCE/DEST 12	2.4	0.5	-0.16	1.6	NA	NA	-30.0	100
SCE/DEST 11	2.4	0.5	-0.16	1.6	NA	NA	-30.0	125
SCE/DEST 10	2.4	0.5	-0.16	1.6	NA	NA	-30.0	125
SCE/DEST 9	2.4	0.5	-0.16	1.6	NA	NA	-30.0	125
SCE/DEST 8	2.4	0.5	-0.16	1.6	NA	NA	-30.0	125
ADRS/WR 15	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75
ADRS/WR 14	2.4	0.5	-0.4	6.0	NA	NA	-100.0	100
ADRS/WR 13	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75
ADRS/WR 12	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75
ADRS/WR 11	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75
ADRS/WR 10	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
MAR 15	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
MAR 14	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
MAR 13	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
MAR 12	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
MAR 11	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
MAR 10	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
MAR 9	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
MAR 8	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
ADRS/WR 9	2.4	0.5	-0.4	6.0	NA	NA	-100.0	125
ADRS/WR 8	2.4	0.5	-0.4	6.0	NA	NA	-100.0	125
ADRS/WR 2	2.4	0.5	-0.4	6.0	NA	NA	-100.0	100
ADRS/WR 3	2.4	0.5	-0.4	6.0	NA	NA	-100.0	100
ADRS/WR 4	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75
ADRS/WR 5	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
ADRS/WR 6	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75
ADRS/WR 7	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75

6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	th	IOC CLK to BSTRP ENBL	14	14	14	-	-	-	nS	12
1	th	IOC CLK to DESTN=CM	14	14	14	-	-	-	nS	12
1	th	IOC CLK to ENBL T1	14	14	14	-	-	-	nS	12
2	th	IOC CLK to DECR (R1)	43	43	43	-	-	-	nS	12
3	th	IOC CLK to IOCA REQ	38	38	38	-	-	-	nS	12
4	th	IOC CLK to CHAN 3	46	46	46	-	-	-	nS	12
4	th	IOC CLK to CHAN 2	46	46	46	-	-	-	nS	12
4	th	IOC CLK to CHAN 1	46	46	46	-	-	-	nS	12
4	th	IOC CLK to CHAN 0	46	46	46	-	-	-	nS	12
4	th	IOC CLK to IOC BUS 1	46	46	46	-	-	-	nS	12
4	th	IOC CLK to CH/INTR HIT	46	46	46	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tsu	ENBL CM WR to IOC CLK	15	15	15	-	-	-	nS	12
6	tsu	DEST CM ADDR to IOC CLK	15	15	15	-	-	-	nS	12
7	tsu	CMMN ADRS 0 to CLK COMM REG	15	15	15	-	-	-	nS	12
7	tsu	CMMN ADRS 1 to CLK COMM REG	15	15	15	-	-	-	nS	12
8	tsu	EI to C4 CLK	3	3	3	-	-	-	nS	12
9	tpd	MAE ENBL to ADRS 7	-	-	-	18	18	18	nS	12
9	tpd	MAE ENBL to ADRS 6	-	-	-	18	18	18	nS	12
9	tpd	MAE ENBL to ADRS 5	-	-	-	18	18	18	nS	12
9	tpd	MAE ENBL to ADRS 4	-	-	-	18	18	18	nS	12
9	tpd	MAE ENBL to ADRS 3	-	-	-	18	18	18	nS	12
9	tpd	MAE ENBL to ADRS 2	-	-	-	18	18	18	nS	12
9	tpd	MAE ENBL to ADRS 1	-	-	-	18	18	18	nS	12
9	tpd	MAE ENBL to ADRS 0	-	-	-	18	18	18	nS	12
10	tpd	ENBL 10-15 to ADRS/WR 15	-	-	-	18	18	18	nS	12
10	tpd	ENBL 10-15 to ADRS/WR 14	-	-	-	18	18	18	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
10	tpd	ENBL 10-15 to ADRS/WR 13	-	-	-	18	18	18	nS	12
10	tpd	ENBL 10-15 to ADRS/WR 12	-	-	-	18	18	18	nS	12
10	tpd	ENBL 10-15 to ADRS/WR 11	-	-	-	18	18	18	nS	12
10	tpd	ENBL 10-15 to ADRS/WR 10	-	-	-	18	18	18	nS	12
11	tpd	ENBL CM/MAR to MAR 5	-	-	-	32	32	32	nS	12
11	tpd	ENBL CM/MAR to MAR 6	-	-	-	32	32	32	nS	12
11	tpd	ENBL CM/MAR to MAR 7	-	-	-	32	32	32	nS	12
11	tpd	ENBL CM/MAR to MAR 2	-	-	-	32	32	32	nS	12
11	tpd	ENBL CM/MAR to MAR 4	-	-	-	32	32	32	nS	12
11	tpd	ENBL CM/MAR to MAR 3	-	-	-	32	32	32	nS	12
11	tpd	ENBL CM/MAR to MAR 1	-	-	-	32	32	32	nS	12
11	tpd	ENBL CM/MAR to MAR 0	-	-	-	32	32	32	nS	12
12	tpd	ENBL ADR 0-9 to ADRS WR 9	-	-	-	18	18	18	nS	12
12	tpd	ENBL ADR 0-9 to ADRS WR 8	-	-	-	18	18	18	nS	12
12	tpd	ENBL ADR 0-9 to ADRS WR 2	-	-	-	18	18	18	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
12	tpd	ENBL ADR 0-9 to ADRS WR 3	-	-	-	18	18	18	nS	12
12	tpd	ENBL ADR 0-9 to ADRS WR 4	-	-	-	18	18	18	nS	12
12	tpd	ENBL ADR 0-9 to ADRS WR 5	-	-	-	18	18	18	nS	12
12	tpd	ENBL ADR 0-9 to ADRS WR 6	-	-	-	18	18	18	nS	12
12	tpd	ENBL ADR 0-9 to ADRS WR 7	-	-	-	18	18	18	nS	12
13	tpd	IOC CLK to BYTE SEL BIT	-	-	-	67	67	67	nS	12

6.4 Module function. The IOC control memory module contains I/O control memory Random Access Memories (RAMs), byte and page control memory RAMs, Buffer Address Pointer (BAP) and Buffer Control Word (BCW) counters, control Memory Address Register (MAR), and the command address register. The I/O control memory RAMs are loaded from the source B/destination bus and store program chaining information for 16 I/O channels. Each I/O channel has 16 storage locations to store data pertinent to an input and output chain. BAP/BCW counters decrement the buffer word count and increment the buffer address pointer during buffer operations. The byte and page control memory RAMs are used to store byte and page select data. The IOC will use this data to store external interrupt and chaining information in main memory by using Memory Address Expansion (MAE) page set 0. If buffer control word bit 13 = 1, MAE page set 2 is used to generate addresses for data transfers on channels 0 through 7 and MAE page set 3 is used to generate addresses for data transfers on channels 8 through 15.

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If bit 13 = 0, MAE page set 0 is selected for all transfers. When the MAE module is not installed, the 16-bit address for chain instructions, external interrupts, and buffer storage is driven from the IOC control memory module. When the MAE module is installed, the control memory MAR receives address information from I/O control memory and places that address information on the common bus for memory reference operations. The command address register holds the address of a Central Processor (CP) command instruction to be executed by the IOC.

6.4.1 Control memory RAMS. I/O control is composed of four 256 x 4-bit random access memory chips. Data is supplied to the control memories by the Source/Destination (S/D) data register when the microinstruction D-field is equal to 01 or 0A. Addressing for the control memories is obtained from two sources:

(a) the source/destination bus via the CM address latch when the microinstruction D-field is equal to 08 or 09.

(b) the data buffer address latch.

Outputs from these latches are sent directly to the control memory RAM address inputs. Write enables are provided by the Enable Control Memory Write signal from the IOC data bus interface (lower) module. Output enables are obtained from an exclusive OR of +5V and External Interrupt (EI) (originating at the IOC data bus interface lower module).

6.4.2 Buffer control word and buffer address counters. The buffer control word and the buffer address pointer are updated by the BCW and BAP counters, respectively. The Buffer Transfer Count (BTC) is decremented by the Dec BCW (dual first pass) signal applied to the count enable inputs to the counters. The BAP is incremented by the Inc BAP signal applied to the BAP counters. Updated BAP and BCW are transferred from the counters and returned back to the addressed control memory locations.

6.4.3 Memory address register. The control memory MAR is a 16-bit register that receives the 16-bit control memory RAM output. Outputs from the MAR are sent to:

(a) the boot detect Programmable Array Logic (PAL) which sends the boot detect signal to the IOC common bus control module and in turn to the address generator and memory address expansion modules.

(b) the common bus via drivers for bits 15-10 and drivers for bits 9-2.

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(c) MAE drivers enabled by the MAE Enable signal from the IOC common bus control module (MAR bits 15-10 with page bits 1 and 0 form MAE address bits 7-0). MAR bits 0 and 1 are placed on the MAR/Memory Write Data (MWD) bus directly. The MAR is controlled by the Enable CM MAR signal from the IOC common bus control module.

6.4.4 Command address register. The command address register is a 16-bit register receiving inputs from the common bus and placing its outputs on the source B/destination bus. Clocking is done by the clock command signal from the IOC common bus control module. Control is provided by the source command signal from the IOC sequencer module. Data supplied to the command address register is the address of a CP command in main memory.

6.4.5 Source/destination bus drivers. The source/destination drivers consist of a series of wire-OR functions which connect the following inputs:

(a) control memory or external interrupt bits 15-0.

(b) command register bits 15-0.

(c) channel control register bits 15-0.

Outputs from the source/destination bus drivers are placed on the source B/destination bus.

6.4.6 S/D control memory address latch and data register. The data buffer address latch receives channel select bits from IOC channel select module as inputs. Outputs are wire-ORed with outputs from the S/D control memory address latch (inputs are source/destination bus bits 7-0) to become the control memory address bus. The source/destination data register receives information from the source B/destination bus, clocked in by clk CM data, and controlled by the IO clock and the dest CM signal from the IOC sequencer module. Outputs from the source/ destination data register are applied via wire-OR inputs (along with BCW/BAP information) to the control memory RAM and byte page control memory RAM data inputs.

6.4.7 Byte and page circuitry. The byte and page circuitry consists of the byte and page multiplexer, the byte and page control memory, and the byte and page register. Inputs to the byte and page multiplexer are page select information and

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byte information. Selection of inputs is accomplished by the destination CM signal from the IOC sequencer module (microinstruction D = 1). Outputs from the byte and page multiplexer are supplied to the byte and page control memory RAM. The byte and page control memory RAM is a 256 x 4 bit RAM. Addressing is obtained from the control memory address bus. Outputs from the byte and page control memory are sent via the byte and page register to the MAE drivers, the bootstrap detector, and also back to the byte and page multiplexer. Outputs are also sent to the source/destination.

6.4.8 Channel control circuits. Channel control circuits consist of the channel register and the channel control latch. The channel register receives channel select information from the IOC channel control module and transfers that data to the channel control register. The channel control register receives the channel information and clocks its outputs to the source/ destination bus.

6.4.9 EI + konstant latch. The external interrupt + konstant latch receives channel select information and IOC select information and places its outputs via the source/ destination bus drivers onto the bus.

6.4.10 Description of I/O pins.

Signal name	Pin number	I/O	Description
IOC CLK	2	I	IOC CLK
DESTN=CM	3	I	DESTN=CM
ENBL CM WR	4	I	ENBL CM WR
BSTRP ENBL	5	I	BOOTSTRAP ENABLE (GR1 BIT 12)
ENBL T1	6	I	ENBL T1
C4 CLK	7	I	C4 CLOCK

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6.4.10 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
BYTE MODE	8	I	BYTE MODE
IOCA REQ	9	I	IOCA SERVICE REQUEST
NDRO BIT MDE	11	I	NDRO BIT MODE
DECR (R1)	12	I	DECR (R1)
SRCE COM REG	13	I	SOURCE COMMAND REGISTER
T4	14	I	T4 X (DMA + ESA)
IOC SEL 0	16	I	IOC SEL BIT 0
IOC SEL 1	17	I	IOC SEL BIT 1
CMMN ADRS 1	19	I	COMMON ADDRESS 1
CMMN ADRS 0	20	I	COMMON ADDRESS 0
CLK COMM REG	30	I	CLOCK COMMAND REG
SRCE CH CTRL	32	I	SOURCE CHANNEL CONTROL
SRCE CM	33	I	SOURCE CM
CHAN 3	34	I	CHANNEL SEL 3
CHAN 2	35	I	CHANNEL SEL 2
CHAN 1	36	I	CHANNEL SEL 1
CHAN 0	37	I	CHANNEL SEL 0
IOC BUS 1	38	I	IOC BUS BIT 1
CH/INTR HIT	39	I	CHAIN/INTRPT HIT + INIT SCAN
ENBL CM/MAR	45	I	ENBL CM MAR

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6.4.10 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
DEST CM ADDR	52	I	DESTN CM ADDRESS
ENBL 10-15	62	I	ENABLE BITS 10-15 (CM)
MAE ENBL	88	I	MAE ENABLE
FRCE P 00	92	I	FORCE PAGE 00
EI	93	I	EXTERNAL INTERRUPT
ENBL ADR 0-9	94	I	ENABLE ADRS (0-9) CB
CLK MAR	15	O	CLK MAR
BOOT DTCT	18	O	BOOT DETECT
BYTE SEL BIT	41	O	BYTE SEL BIT
MAR 0	46	O	MAR (CM) BIT 0
MAR 1	47	O	MAR (CM) BIT 1
DEC BCW	69	O	DEC BCW (R1)
ADRS 7	79	O	ADDRESS BIT 7 (MAE)
ADRS 6	80	O	ADDRESS BIT 6 (MAE)
ADRS 5	82	O	ADDRESS BIT 5 (MAE)
ADRS 4	83	O	ADDRESS BIT 4 (MAE)
ADRS 0	84	O	ADDRESS BIT 0 (MAE)
ADRS 1	85	O	ADDRESS BIT 1 (MAE)
ADRS 2	86	O	ADDRESS BIT 2 (MAE)
ADRS 3	87	O	ADDRESS BIT 3 (MAE)

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6.4.10 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
SCE/DEST 0	21	I/O	IOC SCE/DESTN BUS BIT 0
SCE/DEST 4	22	I/O	IOC SCE/DESTN BUS BIT 4
SCE/DEST 5	23	I/O	IOC SCE/DESTN BUS BIT 5
SCE/DEST 1	24	I/O	IOC SCE/DESTN BUS BIT 1
SCE/DEST 3	26	I/O	IOC SCE/DESTN BUS BIT 3
SCE/DEST 6	27	I/O	IOC SCE/DESTN BUS BIT 6
SCE/DEST 7	28	I/O	IOC SCE/DESTN BUS BIT 7
SCE/DEST 2	29	I/O	IOC SCE/DESTN BUS BIT 2
MAR 5	42	I/O	MAR (CM) BIT 5
MAR 6	43	I/O	MAR (CM) BIT 6
MAR 7	44	I/O	MAR (CM) BIT 7
MAR 2	48	I/O	MAR (CM) BIT 2
MAR 4	49	I/O	MAR (CM) BIT 4
MAR 3	50	I/O	MAR (CM) BIT 3
SCE/DEST 15	53	I/O	IOC SCE/DESTN BUS BIT 15
SCE/DEST 14	54	I/O	IOC SCE/DESTN BUS BIT 14
SCE/DEST 13	55	I/O	IOC SCE/DESTN BUS BIT 13
SCE/DEST 12	56	I/O	IOC SCE/DESTN BUS BIT 12
SCE/DEST 11	57	I/O	IOC SCE/DESTN BUS BIT 11
SCE/DEST 10	58	I/O	IOC SCE/DESTN BUS BIT 10

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6.4.10 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
SCE/DEST 9	59	I/O	IOC SCE/DESTN BUS BIT 9
SCE/DEST 8	61	I/O	IOC SCE/DESTN BUS BIT 8
ADRS/WR 15	63	I/O	CB ADRS/WRITE DATA BIT 15
ADRS/WR 14	64	I/O	CB ADRS/WRITE DATA BIT 14
ADRS/WR 13	65	I/O	CB ADRS/WRITE DATA BIT 13
ADRS/WR 12	66	I/O	CB ADRS/WRITE DATA BIT 12
ADRS/WR 11	67	I/O	CB ADRS/WRITE DATA BIT 11
ADRS/WR 10	68	I/O	CB ADRS/WRITE DATA BIT 10
MAR 15	70	I/O	MAR (CM) BIT 15
MAR 14	71	I/O	MAR (CM) BIT 14
MAR 13	72	I/O	MAR (CM) BIT 13
MAR 12	73	I/O	MAR (CM) BIT 12
MAR 11	74	I/O	MAR (CM) BIT 11
MAR 10	76	I/O	MAR (CM) BIT 10
MAR 9	77	I/O	MAR (CM) BIT 9
MAR 8	78	I/O	MAR (CM) BIT 8
ADRS/WR 9	89	I/O	CB ADRS/WRITE DATA BIT 9
ADRS/WR 8	91	I/O	CB ADRS/WRITE DATA BIT 8
ADRS/WR 2	95	I/O	CB ADRS/WRITE DATA BIT 2
ADRS/WR 3	96	I/O	CB ADRS/WRITE DATA BIT 3

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6.4.10 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
ADRS/WR 4	97	I/O	CB ADRS/WRITE DATA BIT 4
ADRS/WR 5	98	I/O	CB ADRS/WRITE DATA BIT 5
ADRS/WR 6	99	I/O	CB ADRS/WRITE DATA BIT 6
ADRS/WR 7	100	I/O	CB ADRS/WRITE DATA BIT 7

6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N719)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect		25 ± 5°C Initial limits		Units	Notes
		A to	B to	Min	Max		
A	Isolation, frame	C	Frame	10	INF	MOhm	1-8

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A
Isolation measurement	NA	± 500 KOhm	A

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

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NOTES: - Continued.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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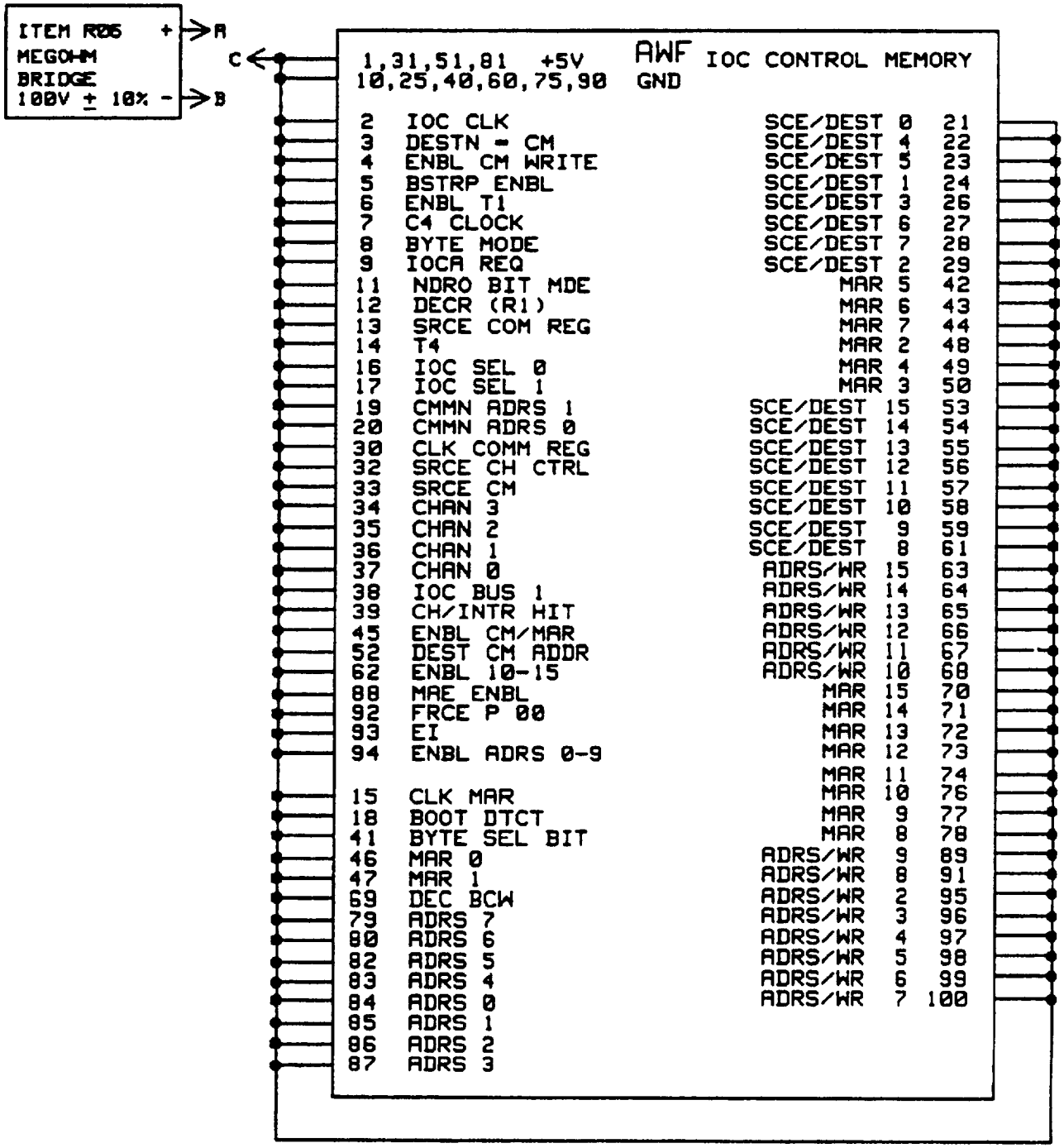


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to	Connect B to	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, +5V	1	31				
B6	Continuity, +5V	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B6

2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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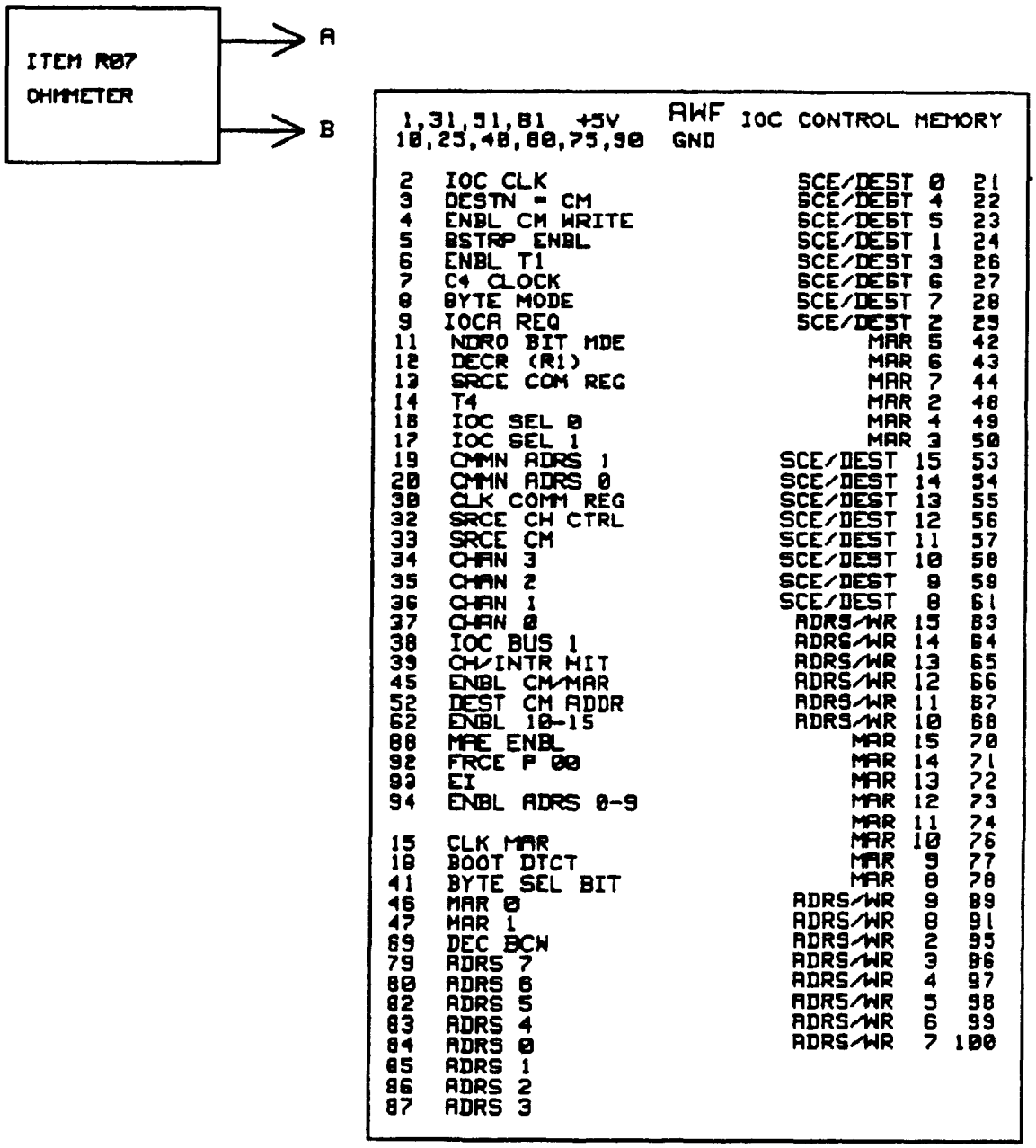


FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect		25 ± 5°C Initial limits	Notes
			Positive lead to	Negative lead to		
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	2.95 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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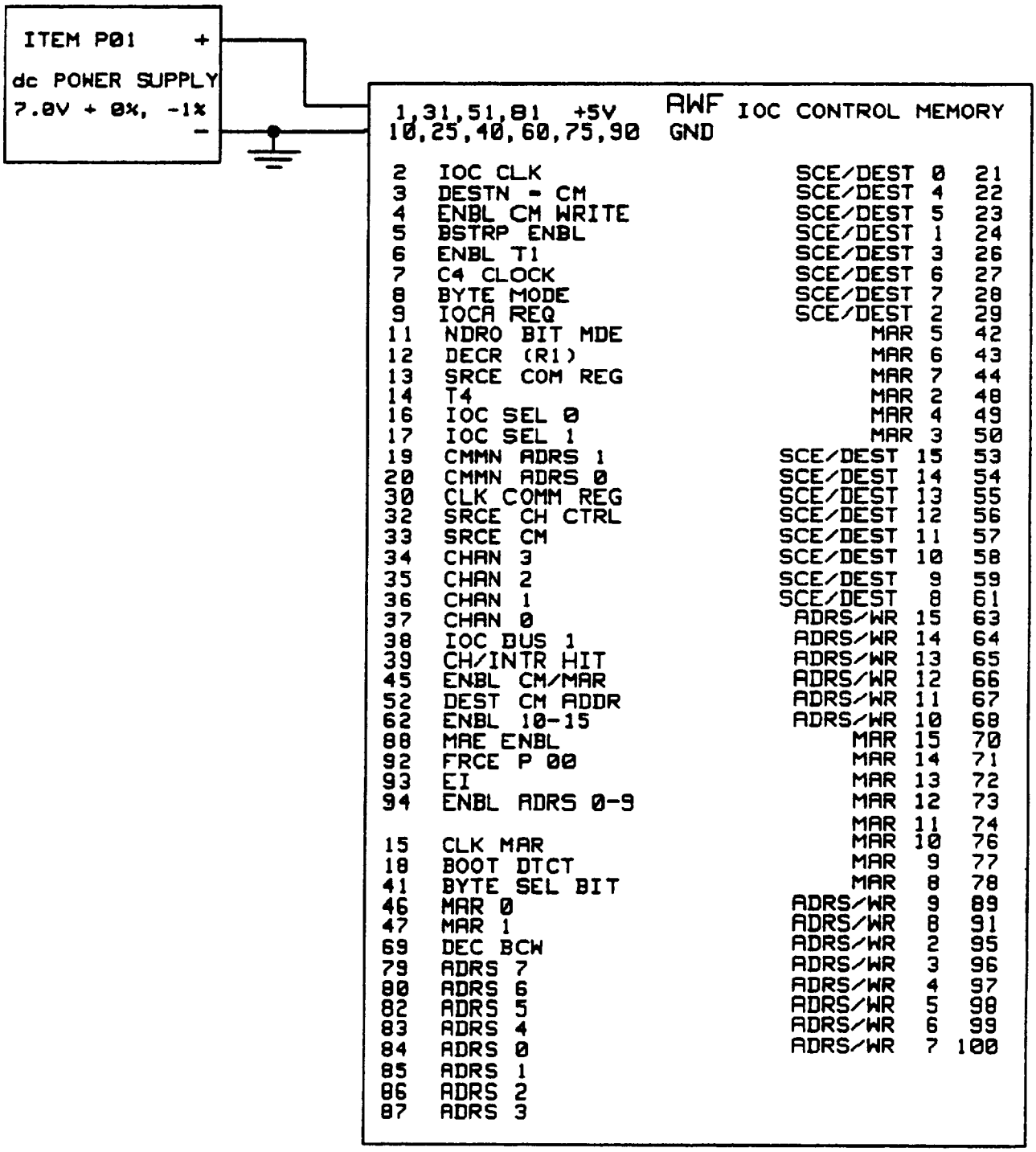


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	IOC CLK	2	5.5	0.0	4.0	mA	1-3
D2	DESTN=CM	3	5.5	0.0	0.2	mA	1-3
D3	ENBL CM WR	4	5.5	0.0	0.1	mA	1-3
D4	BSTRP ENBL	5	5.5	0.0	0.2	mA	1-3
D5	ENBL T1	6	5.5	0.0	0.2	mA	1-3
D6	C4 CLK	7	5.5	0.0	0.1	mA	1-3
D7	BYTE MODE	8	5.5	0.0	1.0	mA	1-3
D8	IOCA REQ	9	5.5	0.0	0.1	mA	1-3
D9	NDRO BIT MDE	11	5.5	0.0	1.1	mA	1-3
D10	DECR (R1)	12	5.5	0.0	0.1	mA	1-3
D11	SRCE COM REG	13	5.5	0.0	0.2	mA	1-3
D12	T4	14	5.5	0.0	1.0	mA	1-3
D13	IOC SEL 0	16	5.5	0.0	0.2	mA	1-3
D14	IOC SEL 1	17	5.5	0.0	0.2	mA	1-3
D15	CMMN ADRS 1	19	5.5	0.0	0.1	mA	1-3
D16	CMMN ADRS 0	20	5.5	0.0	0.1	mA	1-3
D17	CLK COMM REG	30	5.5	0.0	0.2	mA	1-3
D18	SRCE CH CTRL	32	↓	↓	↓	↓	↓
D19	SRCE CM	33	5.5	0.0	0.2	mA	1-3
D20	CHAN 3	34	5.5	0.0	0.3	mA	1-3
D21	CHAN 2	35	5.5	0.0	0.3	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D22	CHAN 1	36	5.5	0.0	0.3	mA	1-3
D23	CHAN 0	37	5.5	0.0	0.3	mA	1-3
D24	IOC BUS 1	38	5.5	0.0	0.1	mA	1-3
D25	CH/INTR HIT	39	5.5	0.0	1.2	mA	1-3
D26	ENBL CM/MAR	45	5.5	0.0	0.2	mA	1-3
D27	DEST CM ADDR	52	5.5	0.0	0.1	mA	1-3
D28	ENBL 10-15	62	5.5	0.0	0.2	mA	1-3
D29	MAE ENBL	88	5.5	0.0	0.2	mA	1-3
D30	FRCE P 00	92	5.5	0.0	2.0	mA	1-3
D31	EI	93	5.5	0.0	0.3	mA	1-3
D32	ENBL ADR 0-9	94	5.5	0.0	0.2	mA	1-3

NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D32
Power supply current limit	2.46 A	NA	D1-D32
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D32

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	D1-D32
low, (VL)	0.0 V	± 100 mV	D1-D32
Input current measurement	NA	± 200 uA	D1
	NA	± 10 uA	D2
	NA	± 5 uA	D3
	NA	± 10 uA	D4,D5
	NA	± 5 uA	D6
	NA	± 50 uA	D7
	NA	± 5 uA	D8
	NA	± 55 uA	D9
	NA	± 5 uA	D10
	NA	± 10 uA	D11
	NA	± 50 uA	D12
	NA	± 10 uA	D13,D14
	NA	± 5 uA	D15,D16
	NA	± 10 uA	D17-D19
	NA	± 15 uA	D20-D23
	NA	± 5 uA	D24
	NA	± 60 uA	D25
	NA	± 10 uA	D26
	NA	± 5 uA	D27
	NA	± 10 uA	D28,D29
	NA	± 100 uA	D30
	NA	± 15 uA	D31
	NA	± 10 uA	D32

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	CLK MAR	15	0.0	-100.0	-40.0	mA	1-4
E2	BOOT DTCT	18	0.0	-150.0	-60.0	mA	1-4
E3	SCE/DEST 0	21	0.0	-260.0	-30.0	mA	1-4
E4	SCE/DEST 0	21	0.0	-220.0	-30.0	mA	1-4
E5	SCE/DEST 4	22	0.0	-260.0	-30.0	mA	1-4
E6	SCE/DEST 4	22	0.0	-220.0	-30.0	mA	1-4
E7	SCE/DEST 4	22	0.0	-220.0	-30.0	mA	1-4
E8	SCE/DEST 5	23	0.0	-260.0	-30.0	mA	1-4
E9	SCE/DEST 5	23	0.0	-220.0	-30.0	mA	1-4
E10	SCE/DEST 5	23	0.0	-220.0	-30.0	mA	1-4
E11	SCE/DEST 1	24	0.0	-260.0	-30.0	mA	1-4
E12	SCE/DEST 1	24	0.0	-220.0	-30.0	mA	1-4
E13	SCE/DEST 1	24	0.0	-220.0	-30.0	mA	1-4
E14	SCE/DEST 3	26	0.0	-260.0	-30.0	mA	1-4
E15	SCE/DEST 3	26	0.0	-220.0	-30.0	mA	1-4
E16	SCE/DEST 6	27	0.0	-260.0	-30.0	mA	1-4
E17	SCE/DEST 6	27	0.0	-220.0	-30.0	mA	1-4
E18	SCE/DEST 6	27	0.0	-220.0	-30.0	mA	1-4
E19	SCE/DEST 7	28	0.0	-260.0	-30.0	mA	1-4
E20	SCE/DEST 7	28	0.0	-220.0	-30.0	mA	1-4
E21	SCE/DEST 7	28	0.0	-220.0	-30.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E22	SCE/DEST 2	29	0.0	-260.0	-30.0	mA	1-4
E23	SCE/DEST 2	29	0.0	-220.0	-30.0	mA	1-4
E24	SCE/DEST 2	29	0.0	-220.0	-30.0	mA	1-4
E25	BYTE SEL BIT	41	0.0	-260.0	-30.0	mA	1-4
E26	MAR 5	42					
E27	MAR 6	43					
E28	MAR 7	44					
E29	MAR 0	46					
E30	MAR 1	47					
E31	MAR 2	48					
E32	MAR 4	49					
E33	MAR 3	50	0.0	-260.0	-30.0	mA	1-4
E34	SCE/DEST 15	53	0.0	-220.0	-30.0	mA	1-4
E35	SCE/DEST 15	53	0.0	-260.0	-30.0	mA	1-4
E36	SCE/DEST 15	53	0.0	-220.0	-30.0	mA	1-4
E37	SCE/DEST 14	54	0.0	-220.0	-30.0	mA	1-4
E38	SCE/DEST 14	54	0.0	-260.0	-30.0	mA	1-4
E39	SCE/DEST 14	54	0.0	-220.0	-30.0	mA	1-4
E40	SCE/DEST 13	55	0.0	-220.0	-30.0	mA	1-4
E41	SCE/DEST 13	55	0.0	-260.0	-30.0	mA	1-4
E42	SCE/DEST 13	55	0.0	-220.0	-30.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E43	SCE/DEST 12	56	0.0	-220.0	-30.0	mA	1-4
E44	SCE/DEST 12	56	0.0	-260.0	-30.0	mA	1-4
E45	SCE/DEST 12	56	0.0	-220.0	-30.0	mA	1-4
E46	SCE/DEST 11	57	0.0	-220.0	-30.0	mA	1-4
E47	SCE/DEST 11	57	0.0	-260.0	-30.0	mA	1-4
E48	SCE/DEST 11	57	0.0	-220.0	-30.0	mA	1-4
E49	SCE/DEST 10	58	0.0	-220.0	-30.0	mA	1-4
E50	SCE/DEST 10	58	0.0	-260.0	-30.0	mA	1-4
E51	SCE/DEST 10	58	0.0	-220.0	-30.0	mA	1-4
E52	SCE/DEST 9	59	0.0	-220.0	-30.0	mA	1-4
E53	SCE/DEST 9	59	0.0	-260.0	-30.0	mA	1-4
E54	SCE/DEST 9	59	0.0	-220.0	-30.0	mA	1-4
E55	SCE/DEST 8	61	0.0	-220.0	-30.0	mA	1-4
E56	SCE/DEST 8	61	0.0	-260.0	-30.0	mA	1-4
E57	SCE/DEST 8	61	0.0	-220.0	-30.0	mA	1-4
E58	ADRS/WR 15	63	0.0	-325.0	-100.0	mA	1-4
E59	ADRS/WR 14	64					
E60	ADRS/WR 13	65					
E61	ADRS/WR 12	66					
E62	ADRS/WR 11	67					
E63	ADRS/WR 10	68	0.0	-325.0	-100.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E64	DEC BCW	69	0.0	-100.0	-20.0	mA	1-4
E65	MAR 15	70	0.0	-260.0	-30.0	mA	1-4
E66	MAR 14	71					
E67	MAR 13	72					
E68	MAR 12	73					
E69	MAR 11	74					
E70	MAR 10	76					
E71	MAR 9	77					
E72	MAR 8	78	0.0	-260.0	-30.0	mA	1-4
E73	ADRS 7	79	0.0	-325.0	-100.0	mA	1-4
E74	ADRS 6	80					
E75	ADRS 5	82					
E76	ADRS 4	83					
E77	ADRS 0	84					
E78	ADRS 1	85					
E79	ADRS 2	86					
E80	ADRS 3	87					
E81	ADRS/WR 9	89					
E82	ADRS/WR 8	91					
E83	ADRS/WR 2	95					
E84	ADRS/WR 3	96	0.0	-325.0	-100.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E85	ADRS/WR 4	97	0.0	-325.0	-100.0	mA	1-4
E86	ADRS/WR 5	98					
E87	ADRS/WR 6	99					
E88	ADRS/WR 7	100	0.0	-325.0	-100.0	mA	1-4

NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E88
Power supply current limit	2.46 A	NA	E1-E88
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E88
Pattern voltage high, (VH)	3.0 V	± 100 mV	E1-E88
low, (VL)	0.0 V	± 100 mV	E1-E88
Output current measurement	NA	± 3.00 mA	E1
	NA	± 4.50 mA	E2
	NA	± 11.5 mA	E3
	NA	± 9.50 mA	E4
	NA	± 11.5 mA	E5
	NA	± 9.50 mA	E6,E7
	NA	± 11.5 mA	E8

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output current measurement	NA	± 9.50 mA	E9,E10
	NA	± 11.5 mA	E11
	NA	± 9.50 mA	E12,E13
	NA	± 11.5 mA	E14
	NA	± 9.50 mA	E15
	NA	± 11.5 mA	E16
	NA	± 9.50 mA	E17,E18
	NA	± 11.5 mA	E19
	NA	± 9.50 mA	E20,E21
	NA	± 11.5 mA	E22
	NA	± 9.50 mA	E23,E24
	NA	± 11.5 mA	E25-E33
	NA	± 9.50 mA	E34
	NA	± 11.5 mA	E35
	NA	± 9.50 mA	E36,E37
	NA	± 11.5 mA	E38
	NA	± 9.50 mA	E39,E40
	NA	± 11.5 mA	E41
	NA	± 9.50 mA	E42,E43
	NA	± 11.5 mA	E44
	NA	± 9.50 mA	E45,E46
	NA	± 11.5 mA	E47
	NA	± 9.50 mA	E48,E49
	NA	± 11.5 mA	E50
	NA	± 9.50 mA	E51,E52
	NA	± 11.5 mA	E53
	NA	± 9.50 mA	E54,E55
	NA	± 11.5 mA	E56
	NA	± 9.50 mA	E57
	NA	± 11.2 mA	E58-E63
	NA	± 4.00 mA	E64
	NA	± 11.5 mA	E65-E72
	NA	± 11.2 mA	E73-E88

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	IOC CLK	2	2.4	-8.0	0.2	mA	1-3
F2	DESTN=CM	3	2.4	-0.8	0.04	mA	1-3
F3	ENBL CM WR	4	2.4	-0.6	0.02	mA	1-3
F4	BSTRP ENBL	5	2.4	-0.4	0.04	mA	1-3
F5	ENBL T1	6	2.4	-0.4	0.04	mA	1-3
F6	C4 CLK	7	2.4	-0.6	0.02	mA	1-3
F7	BYTE MODE	8	2.4	-2.0	0.05	mA	1-3
F8	IOCA REQ	9	2.4	-0.4	0.02	mA	1-3
F9	NDRO BIT MDE	11	2.4	-0.45	0.05	mA	1-3
F10	DECR (RJ.)	12	2.4	-0.4	0.02	mA	1-3
F11	SRCE COM REG	13	2.4	-0.4	0.04	mA	1-3
F12	T4	14	2.4	-2.0	0.05	mA	1-3
F13	IOC SEL 0	16	2.4	-0.4	0.04	mA	1-3
F14	IOC SEL 1	17	2.4	-0.4	0.04	mA	1-3
F15	CMMN ADRS 1	19	2.4	-0.2	0.02	mA	1-3
F16	CMMN ADRS 0	20	2.4	-0.2	0.02	mA	1-3
F17	SCE/DEST 0	21	2.4	-0.46	0.1	mA	1-3
F18	SCE/DEST 4	22					
F19	SCE/DEST 5	23					
F20	SCE/DEST 1	24					
F21	SCE/DEST 3	26	2.4	-0.46	0.1	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F22	SCE/DEST 6	27	2.4	-0.46	0.1	mA	1-3
F23	SCE/DEST 7	28					
F24	SCE/DEST 2	29	2.4	-0.46	0.1	mA	1-3
F25	CLK COMM REG	30	2.4	-0.4	0.04	mA	1-3
F26	SRCE CH CTRL	32					
F27	SRCE CM	33	2.4	-0.4	0.04	mA	1-3
F28	CHAN 3	34	2.4	-0.6	0.06	mA	1-3
F29	CHAN 2	35					
F30	CHAN 1	36					
F31	CHAN 0	37	2.4	-0.6	0.06	mA	1-3
F32	IOC BUS 1	38	2.4	-0.2	0.02	mA	1-3
F33	CH/INTR HIT	39	2.4	-2.8	0.09	mA	1-3
F34	MAR 5	42	2.4	-1.62	0.04	mA	1-3
F35	MAR 6	43	2.4	-1.87	0.07	mA	1-3
F36	MAR 7	44	2.4	-1.87	0.07	mA	1-3
F37	ENBL CM/MAR	45	2.4	-0.4	0.04	mA	1-3
F38	MAR 2	48	2.4	-1.62	0.04	mA	1-3
F39	MAR 4	49					
F40	MAR 3	50	2.4	-1.62	0.04	mA	1-3
F41	DEST CM ADDR	52	2.4	-0.6	0.02	mA	1-3
F42	SCE/DEST 15	53	2.4	-0.26	0.08	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
F43	SCE/DEST 14	54	2.4	-0.26	0.08	mA	1-3
F44	SCE/DEST 13	55					
F45	SCE/DEST 12	56					
F46	SCE/DEST 11	57					
F47	SCE/DEST 10	58					
F48	SCE/DEST 9	59					
F49	SCE/DEST 8	61	2.4	-0.26	0.08	mA	1-3
F50	ENBL 10-15	62	2.4	-2.0	0.04	mA	1-3
F51	ADRS/WR 15	63	2.4	-0.25	0.07	mA	1-3
F52	ADRS/WR 14	64					
F53	ADRS/WR 13	65					
F54	ADRS/WR 12	66					
F55	ADRS/WR 11	67					
F56	ADRS/WR 10	68	2.4	-0.25	0.07	mA	1-3
F57	MAR 15	70	2.4	-3.47	0.09	mA	1-3
F58	MAR 14	71					
F59	MAR 13	72					
F60	MAR 12	73					
F61	MAR 11	74					
F62	MAR 10	76	2.4	-3.47	0.09	mA	1-3
F63	MAR 9	77	2.4	-1.87	0.07	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F64	MAR 8	78	2.4	-1.87	0.07	mA	1-3
F65	MAE ENBL	88	2.4	-2.0	0.04	mA	1-3
F66	ADRS/WR 9	89	2.4	-0.25	0.07	mA	1-3
F67	ADRS/WR 8	91	2.4	-0.25	0.07	mA	1-3
F68	FRCE P 00	92	2.4	-4.0	0.1	mA	1-3
F69	EI	93	2.4	-1.8	0.06	mA	1-3
F70	ENBL ADR 0-9	94	2.4	-2.0	0.04	mA	1-3
F71	ADRS/WR 2	95	2.4	-0.25	0.07	mA	1-3
F72	ADRS/WR 3	96					
F73	ADRS/WR 4	97					
F74	ADRS/WR 5	98					
F75	ADRS/WR 6	99					
F76	ADRS/WR 7	100	2.4	-0.25	0.07	mA	1-3

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F76
Power supply current limit	2.46 A	NA	F1-F76

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:22Z
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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F76
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	F1-F76
low, (VL)	0.0 V	± 100 mV	F1-F76
Input current measurement	NA	± 410 uA	F1
	NA	± 42 uA	F2
	NA	± 31 uA	F3
	NA	± 22 uA	F4,F5
	NA	± 31 uA	F6
	NA	± 102 uA	F7
	NA	± 21 uA	F8
	NA	± 25 uA	F9
	NA	± 21 uA	F10
	NA	± 22 uA	F11
	NA	± 102 uA	F12
	NA	± 22 uA	F13,F14
	NA	± 11 uA	F15,F16
	NA	± 28 uA	F17-F24
	NA	± 22 uA	F25-F27
	NA	± 33 uA	F28-F31
	NA	± 11 uA	F32
	NA	± 144 uA	F33
	NA	± 83 uA	F34
	NA	± 97 uA	F35,F36
	NA	± 22 uA	F37
	NA	± 83 uA	F38-F40
	NA	± 31 uA	F41
	NA	± 17 uA	F42-F49
	NA	± 102 uA	F50
	NA	± 16 uA	F51-F56
	NA	± 178 uA	F57-F62
	NA	± 97 uA	F63,F64
	NA	± 102 uA	F65
	NA	± 16 uA	F66,F67
	NA	± 205 uA	F68

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA NA NA	± 93 uA ± 102 uA ± 16 uA	F69 F70 F71-F76

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	IOC CLK	2	0.4	-8.0	0.2	mA	1-3
G2	DESTN=CM	3	0.4	-0.8	0.04	mA	1-3
G3	ENBL CM WR	4	0.4	-0.6	0.02	mA	1-3
G4	BSTRP ENBL	5	0.4	-0.4	0.04	mA	1-3
G5	ENBL T1	6	0.4	-0.4	0.04	mA	1-3
G6	C4 CLK	7	0.4	-0.6	0.02	mA	1-3
G7	BYTE MODE	8	0.4	-2.0	0.05	mA	1-3
G8	IOCA REQ	9	0.4	-0.4	0.02	mA	1-3
G9	NDRO BIT MDE	11	0.4	-0.45	0.05	mA	1-3
G10	DECR (R1)	12	0.4	-0.4	0.02	mA	1-3
G11	SRCE COM REG	13	0.4	-0.4	0.04	mA	1-3
G12	T4	14	0.4	-2.0	0.05	mA	1-3
G13	IOC SEL 0	16	0.4	-0.4	0.04	mA	1-3
G14	IOC SEL 1	17	0.4	-0.4	0.04	mA	1-3
G15	CMMN ADRS 1	19	0.4	-0.2	0.02	mA	1-3
G16	CMMN ADRS 0	20	0.4	-0.2	0.02	mA	1-3
G17	SCE/DEST 0	21	0.4	-0.46	0.1	mA	1-3
G18	SCE/DEST 4	22					
G19	SCE/DEST 5	23					
G20	SCE/DEST 1	24					
G21	SCE/DEST 3	26	0.4	-0.46	0.1	mA	1-3

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Check the source to verify that this is the current version before use.

TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G22	SCE/DEST 6	27	0.4	-0.46	0.1	mA	1-3
G23	SCE/DEST 7	28					
G24	SCE/DEST 2	29	0.4	-0.46	0.1	mA	1-3
G25	CLK COMM REG	30	0.4	-0.4	0.04	mA	1-3
G26	SRCE CH CTRL	32					
G27	SRCE CM	33	0.4	-0.4	0.04	mA	1-3
G28	CHAN 3	34	0.4	-0.6	0.06	mA	1-3
G29	CHAN 2	35					
G30	CHAN 1	36					
G31	CHAN 0	37	0.4	-0.6	0.06	mA	1-3
G32	IOC BUS 1	38	0.4	-0.2	0.02	mA	1-3
G33	CH/INTR HIT	39	0.4	-2.8	0.09	mA	1-3
G34	MAR 5	42	0.4	-1.62	0.04	mA	1-3
G35	MAR 6	43	0.4	-1.87	0.07	mA	1-3
G36	MAR 7	44	0.4	-1.87	0.07	mA	1-3
G37	ENBL CM/MAR	45	0.4	-0.4	0.04	mA	1-3
G38	MAR 2	48	0.4	-1.62	0.04	mA	1-3
G39	MAR 4	49					
G40	MAR 3	50	0.4	-1.62	0.04	mA	1-3
G41	DEST CM ADDR	52	0.4	-0.6	0.02	mA	1-3
G42	SCE/DEST 15	53	0.4	-0.26	0.08	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G43	SCE/DEST 14	54	0.4	-0.26	0.08	mA	1-3
G44	SCE/DEST 13	55					
G45	SCE/DEST 12	56					
G46	SCE/DEST 11	57					
G47	SCE/DEST 10	58					
G48	SCE/DEST 9	59					
G49	SCE/DEST 8	61	0.4	-0.26	0.08	mA	1-3
G50	ENBL 10-15	62	0.4	-2.0	0.04	mA	1-3
G51	ADRS/WR 15	63	0.4	-0.25	0.07	mA	1-3
G52	ADRS/WR 14	64					
G53	ADRS/WR 13	65					
G54	ADRS/WR 12	66					
G55	ADRS/WR 11	67					
G56	ADRS/WR 10	68	0.4	-0.25	0.07	mA	1-3
G57	MAR 15	70	0.4	-3.47	0.09	mA	1-3
G58	MAR 14	71					
G59	MAR 13	72					
G60	MAR 12	73					
G61	MAR 11	74					
G62	MAR 10	76	0.4	-3.47	0.09	mA	1-3
G63	MAR 9	77	0.4	-1.87	0.07	mA	1-3

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Check the source to verify that this is the current version before use.

TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G64	MAR 8	78	0.4	-1.87	0.07	mA	1-3
G65	MAE ENBL	88	0.4	-2.0	0.04	mA	1-3
G66	ADRS/WR 9	89	0.4	-0.25	0.07	mA	1-3
G67	ADRS/WR 8	91	0.4	-0.25	0.07	mA	1-3
G68	FRCE P 00	92	0.4	-4.0	0.1	mA	1-3
G69	EI	93	0.4	-1.8	0.06	mA	1-3
G70	ENBL ADR 0-9	94	0.4	-2.0	0.04	mA	1-3
G71	ADRS/WR 2	95	0.4	-0.25	0.07	mA	1-3
G72	ADRS/WR 3	96					
G73	ADRS/WR 4	97					
G74	ADRS/WR 5	98					
G75	ADRS/WR 6	99					
G76	ARDS/WR 7	100	0.4	-0.25	0.07	mA	1-3

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G76
Power supply current limit	2.46 A	NA	G1-G76

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:22Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G76
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 G1-G76 G1-G76
Input current measurement	NA	± 410 uA	G1
	NA	± 42 uA	G2
	NA	± 31 uA	G3
	NA	± 22 uA	G4,G5
	NA	± 31 uA	G6
	NA	± 102 uA	G7
	NA	± 21 uA	G8
	NA	± 25 uA	G9
	NA	± 21 uA	G10
	NA	± 22 uA	G11
	NA	± 102 uA	G12
	NA	± 22 uA	G13,G14
	NA	± 11 uA	G15,G16
	NA	± 28 uA	G17-G24
	NA	± 22 uA	G25-G27
	NA	± 33 uA	G28-G31
	NA	± 11 uA	G32
	NA	± 144 uA	G33
	NA	± 83 uA	G34
	NA	± 97 uA	G35,G36
	NA	± 22 uA	G37
	NA	± 83 uA	G38-G40
	NA	± 31 uA	G41
	NA	± 17 uA	G42-G49
	NA	± 102 uA	G50
	NA	± 16 uA	G51-G56
	NA	± 178 uA	G57-G62
	NA	± 97 uA	G63,G64
	NA	± 102 uA	G65
	NA	± 16 uA	G66,G67
	NA	± 205 uA	G68

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA NA NA	± 93 uA ± 102 uA ± 16 uA	G69 G70 G71-G76

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:22Z
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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	CLK MAR	15	-0.4	2.4	4.5	V	1-4
H2	BOOT DTCT	18	-0.4	2.4	4.5	V	1-4
H3	SCE/DEST 0	21	-0.16	2.4	4.5	V	1-4
H4	SCE/DEST 0	21					
H5	SCE/DEST 4	22					
H6	SCE/DEST 4	22					
H7	SCE/DEST 4	22					
H8	SCE/DEST 5	23					
H9	SCE/DEST 5	23					
H10	SCE/DEST 5	23					
H11	SCE/DEST 1	24					
H12	SCE/DEST 1	24					
H13	SCE/DEST 1	24					
H14	SCE/DEST 3	26					
H15	SCE/DEST 3	26					
H16	SCE/DEST 6	27					
H17	SCE/DEST 6	27					
H18	SCE/DEST 6	27					
H19	SCE/DEST 7	28					
H20	SCE/DEST 7	28					
H21	SCE/DEST 7	28	-0.16	2.4	4.5	V	1-4

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Check the source to verify that this is the current version before use.

TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H22	SCE/DEST 2	29	-0.16	2.4	4.5	V	1-4
H23	SCE/DEST 2	29					
H24	SCE/DEST 2	29					
H25	BYTE SEL BIT	41					
H26	MAR 5	42					
H27	MAR 6	43					
H28	MAR 7	44					
H29	MAR 0	46					
H30	MAR 1	47					
H31	MAR 2	48					
H32	MAR 4	49					
H33	MAR 3	50					
H34	SCE/DEST 15	53					
H35	SCE/DEST 15	53					
H36	SCE/DEST 15	53					
H37	SCE/DEST 14	54					
H38	SCE/DEST 14	54					
H39	SCE/DEST 14	54					
H40	SCE/DEST 13	55					
H41	SCE/DEST 13	55					
H42	SCE/DEST 13	55	-0.16	2.4	4.5	V	1-4

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Check the source to verify that this is the current version before use.

TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H43	SCE/DEST 12	56	-0.16	2.4	4.5	V	1-4
H44	SCE/DEST 12	56					
H45	SCE/DEST 12	56					
H46	SCE/DEST 11	57					
H47	SCE/DEST 11	57					
H48	SCE/DEST 11	57					
H49	SCE/DEST 10	58					
H50	SCE/DEST 10	58					
H51	SCE/DEST 10	58					
H52	SCE/DEST 9	59					
H53	SCE/DEST 9	59					
H54	SCE/DEST 9	59					
H55	SCE/DEST 8	61					
H56	SCE/DEST 8	61					
H57	SCE/DEST 8	61	-0.16	2.4	4.5	V	1-4
H58	ADRS/WR 15	63	-0.4	2.4	4.5	V	1-4
H59	ADRS/WR 14	64					
H60	ADRS/WR 13	65					
H61	ADRS/WR 12	66					
H62	ADRS/WR 11	67					
H63	ADRS/WR 10	68	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H64	DEC BCW	69	-0.16	2.4	4.5	V	1-4
H65	MAR 15	70					
H66	MAR 14	71					
H67	MAR 13	72					
H68	MAR 12	73					
H69	MAR 11	74					
H70	MAR 10	76					
H71	MAR 9	77					
H72	MAR 8	78	-0.16	2.4	4.5	V	1-4
H73	ADRS 7	79	-0.4	2.4	4.5	V	1-4
H74	ADRS 6	80					
H75	ADRS 5	82					
H76	ADRS 4	83					
H77	ADRS 0	84					
H78	ADRS 1	85					
H79	ADRS 2	86					
H80	ADRS 3	87					
H81	ADRS/WR 9	89					
H82	ADRS/WR 8	91					
H83	ADRS/WR 2	95					
H84	ADRS/WR 3	96	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H85	ADRS/WR 4	97	-0.4	2.4	4.5	V	1-4
H86	ADRS/WR 5	98					
H87	ADRS/WR 6	99					
H88	ADRS/WR 7	100	-0.4	2.4	4.5	V	1-4

NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H88
Power supply current limit	2.46 A	NA	H1-H88
Forcing current, (Ioh)	-0.4 mA	± 40 uA	H1,H2
	-0.16 mA	± 16 uA	H3-H57
	-0.4 mA	± 40 uA	H58-H63
	-0.16 mA	± 16 uA	H64-H72
	-0.4 mA	± 40 uA	H73-H88
Pattern voltage high, (VH)	3.0 V	± 100 mV	H1-H88
low, (VL)	0.0 V	± 100 mV	H1-H88
Output voltage measurement	NA	± 105 mV	H1-H88

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

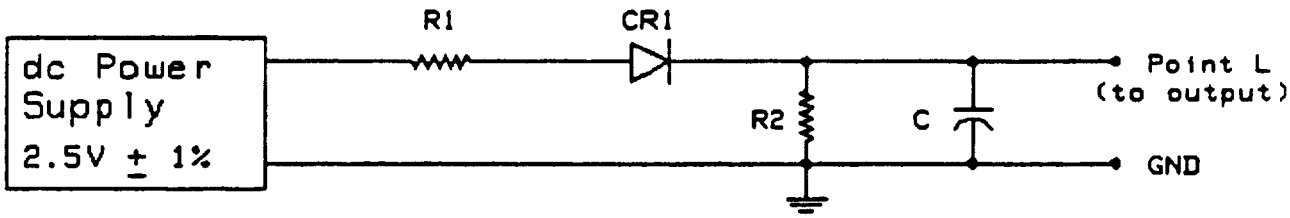
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
15,18,63,64,65, 66,67,68,79,80, 82,83,84,85,86, 87,89,91,95,96, 97,98,99,100	6200	6.0	0.5
21,22,23,24,26, 27,28,29,41,42, 43,44,46,47,48, 49,50,53,54,55, 56,57,58,59,61, 69,70,71,72,73, 74,76,77,78	16000	1.6	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	CLK MAR	15	6.0	0.0	0.5	V	1-4
J2	BOOT DTCT	18	6.0	0.0	0.5	V	1-4
J3	SCE/DEST 0	21	1.6	0.0	0.5	V	1-4
J4	SCE/DEST 0	21					
J5	SCE/DEST 0	21					
J6	SCE/DEST 4	22					
J7	SCE/DEST 4	22					
J8	SCE/DEST 4	22					
J9	SCE/DEST 5	23					
J10	SCE/DEST 5	23					
J11	SCE/DEST 5	23					
J12	SCE/DEST 1	24					
J13	SCE/DEST 1	24					
J14	SCE/DEST 3	26					
J15	SCE/DEST 3	26					
J16	SCE/DEST 3	26					
J17	SCE/DEST 6	27					
J18	SCE/DEST 6	27					
J19	SCE/DEST 6	27					
J20	SCE/DEST 7	28					
J21	SCE/DEST 7	28	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J22	SCE/DEST 7	28	1.6	0.0	0.5	V	1-4
J23	SCE/DEST 2	29					
J24	SCE/DEST 2	29					
J25	SCE/DEST 2	29					
J26	BYTE SEL BIT	41					
J27	MAR 5	42					
J28	MAR 6	43					
J29	MAR 7	44					
J30	MAR 0	46					
J31	MAR 1	47					
J32	MAR 2	48					
J33	MAR 4	49					
J34	MAR 3	50					
J35	SCE/DEST 15	53					
J36	SCE/DEST 15	53					
J37	SCE/DEST 15	53					
J38	SCE/DEST 14	54					
J39	SCE/DEST 14	54					
J40	SCE/DEST 14	54					
J41	SCE/DEST 13	55					
J42	SCE/DEST 13	55	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J43	SCE/DEST 13	55	1.6	0.0	0.5	V	1-4
J44	SCE/DEST 12	56					
J45	SCE/DEST 12	56					
J46	SCE/DEST 12	56					
J47	SCE/DEST 11	57					
J48	SCE/DEST 11	57					
J49	SCE/DEST 11	57					
J50	SCE/DEST 10	58					
J51	SCE/DEST 10	58					
J52	SCE/DEST 10	58					
J53	SCE/DEST 9	59					
J54	SCE/DEST 9	59					
J55	SCE/DEST 9	59					
J56	SCE/DEST 8	61					
J57	SCE/DEST 8	61					
J58	SCE/DEST 8	61	1.6	0.0	0.5	V	1-4
J59	ADRS/WR 15	63	6.0	0.0	0.5	V	1-4
J60	ADRS/WR 14	64					
J61	ADRS/WR 13	65					
J62	ADRS/WR 12	66					
J63	ADRS/WR 11	67	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J64	ADRS/WR 10	68	6.0	0.0	0.5	V	1-4
J65	DEC BCW	69	1.6	0.0	0.5	V	1-4
J66	MAR 15	70					
J67	MAR 14	71					
J68	MAR 13	72					
J69	MAR 12	73					
J70	MAR 11	74					
J71	MAR 10	76					
J72	MAR 9	77					
J73	MAR 8	78	1.6	0.0	0.5	V	1-4
J74	ADRS 7	79	6.0	0.0	0.5	V	1-4
J75	ADRS 6	80					
J76	ADRS 5	82					
J77	ADRS 4	83					
J78	ADRS 0	84					
J79	ADRS 1	85					
J80	ADRS 2	86					
J81	ADRS 3	87					
J82	ADRS/WR 9	89					
J83	ADRS/WR 8	91					
J84	ADRS/WR 2	95	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J85	ADRS/WR 3	96	6.0	0.0	0.5	V	1-4
J86	ADRS/WR 4	97					
J87	ADRS/WR 5	98					
J88	ADRS/WR 6	99					
J89	ADRS/WR 7	100	6.0	0.0	0.5	V	1-4

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J89
Power supply current limit	2.46 A	NA	J1-J89
Forcing current, (Iol)	6.0 mA	± 600 uA	J1,J2
	1.6 mA	± 160 uA	J3-J58
	6.0 mA	± 600 uA	J59-J64
	1.6 mA	± 160 uA	J65-J73
	6.0 mA	± 600 uA	J74-J89
Pattern voltage high, (VH)	3.0 V	± 100 mV	J1-J89
low, (VL)	0.0 V	± 100 mV	J1-J89

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage measurement	NA	± 25 mV	J1-J89

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
High level							
K1	MAR 0	46	2.4	-0.02	0.02	mA	1-3
K2	MAR 1	47	2.4	-0.02	0.02	mA	1-3
K3	ADRS 7	79	2.4	-0.05	0.05	mA	1-3
K4	ADRS 6	80					
K5	ADRS 5	82					
K6	ADRS 4	83					
K7	ADRS 0	84					
K8	ADRS 1	85					
K9	ADRS 2	86					
K10	ADRS 3	87	2.4	-0.05	0.05	mA	1-3
Low level							
K11	MAR 0	46	0.4	-0.02	0.02	mA	1-3
K12	MAR 1	47	0.4	-0.02	0.02	mA	1-3
K13	ADRS 7	79	0.4	-0.05	0.05	mA	1-3
K14	ADRS 6	80					
K15	ADRS 5	82					
K16	ADRS 4	83					
K17	ADRS 0	84					
K18	ADRS 1	85	0.4	-0.05	0.05	mA	1-3

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TABLE XI. Output leakage current, high and low level, test - Continued.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
K19	ADRS 2	86	0.4	-0.05	0.05	mA	1-3
K20	ADRS 3	87	0.4	-0.05	0.05	mA	1-3

NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K1-K20
Power supply current limit	2.46 A	NA	K1-K20
Forcing voltage, (Voh) (Vol)	2.4 V 0.4 V	± 50 mV ± 50 mV	K1-K10 K11-K20
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	K1-K20 K1-K20
Output current measurement	NA NA NA NA	± 2 uA ± 5 uA ± 2 uA ± 5 uA	K1,K2 K3-K10 K11,K12 K13-K20

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L1	IOC CLK	2	-18.0	-1.2	0.0	V	1-3
L2	DESTN=CM	3					
L3	ENBL CM WR	4	-18.0	-1.2	0.0	V	1-3
L4	BSTRP ENBL	5	-18.0	-1.5	0.0	V	1-3
L5	ENBL T1	6	-18.0	-1.5	0.0	V	1-3
L6	C4 CLK	7	-18.0	-1.2	0.0	V	1-3
L7	BYTE MODE	8	-18.0	-1.2	0.0	V	1-3
L8	IOCA REQ	9	-18.0	-1.5	0.0	V	1-3
L9	NDRO BIT MDE	11					
L10	DECR (R1)	12					
L11	SRCE COM REG	13	-18.0	-1.5	0.0	V	1-3
L12	T4	14	-18.0	-1.2	0.0	V	1-3
L13	IOC SEL 0	16	-18.0	-1.5	0.0	V	1-3
L14	IOC SEL 1	17					
L15	CMMN ADRS 1	19					
L16	CMMN ADRS 0	20					
L17	CLK COMM REG	30					
L18	SRCE CH CTRL	32					
L19	SRCE CM	33					
L20	CHAN 3	34					
L21	CHAN 2	35	-18.0	-1.5	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L22	CHAN 1	36	-18.0	-1.5	0.0	V	1-3
L23	CHAN 0	37					
L24	IOC BUS 1	38	-18.0	-1.5	0.0	V	1-3
L25	CH/INTR HIT	39	-18.0	-1.2	0.0	V	1-3
L26	ENBL CM/MAR	45	-18.0	-1.5	0.0	V	1-3
L27	DEST CM ADDR	52	-18.0	-1.2	0.0	V	1-3
L28	ENBL 10-15	62					
L29	MAE ENBL	88					
L30	FRCE P 00	92					
L31	EI	93					
L32	ENBL ADR 0-9	94	-18.0	-1.2	0.0	V	1-3

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L32
Power supply current limit	NA	NA	L1-L32
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L32

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L32 L1-L32
Input voltage measurement	NA NA NA NA NA NA NA NA NA	± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV	L1-L3 L4,L5 L6,L7 L8-L11 L12 L13-L24 L25 L26 L27-L32

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	5.9	11.0	uF	1-4
M2	PSFC	51	6.4	12.0	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA NA	± 255 nF ± 280 nF	M1 M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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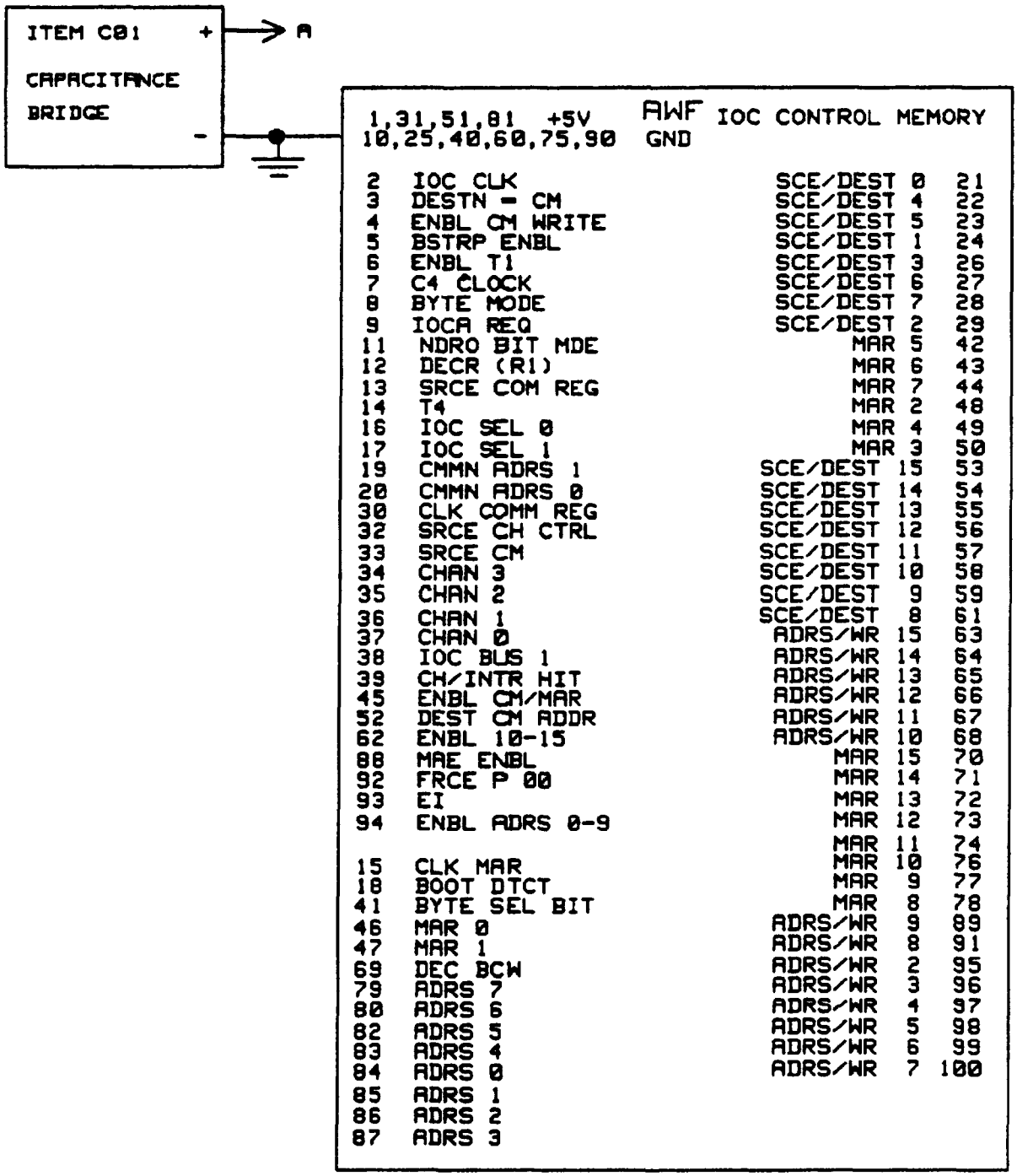


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31, 51,81	640	1637	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	2.46 A	NA	N
Pattern voltage high, (VH)	3.0 V	± 100 mV	N
low, (VL)	0.0 V	± 100 mV	N
Power supply current measurement	NA	± 49.8 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes	Figs
P1	Functional test minimum power supply voltage maximum pattern rate (425 nS)	1-3	9,11
P2	Functional test maximum power supply voltage maximum pattern rate (425 nS)	1-3	9,11

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	2.46 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 25 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

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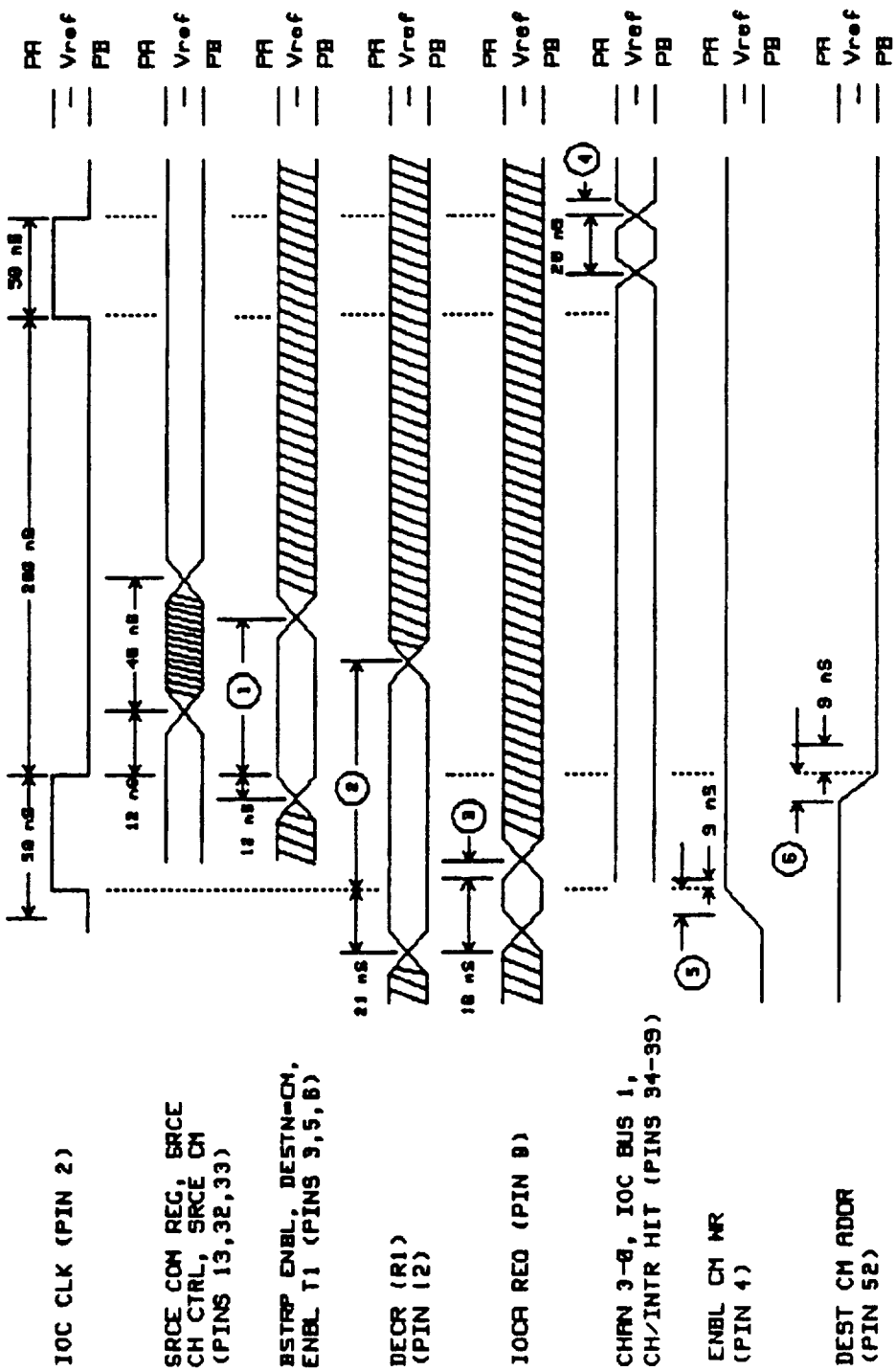


FIGURE 11a. Input timing parameters for table XV.

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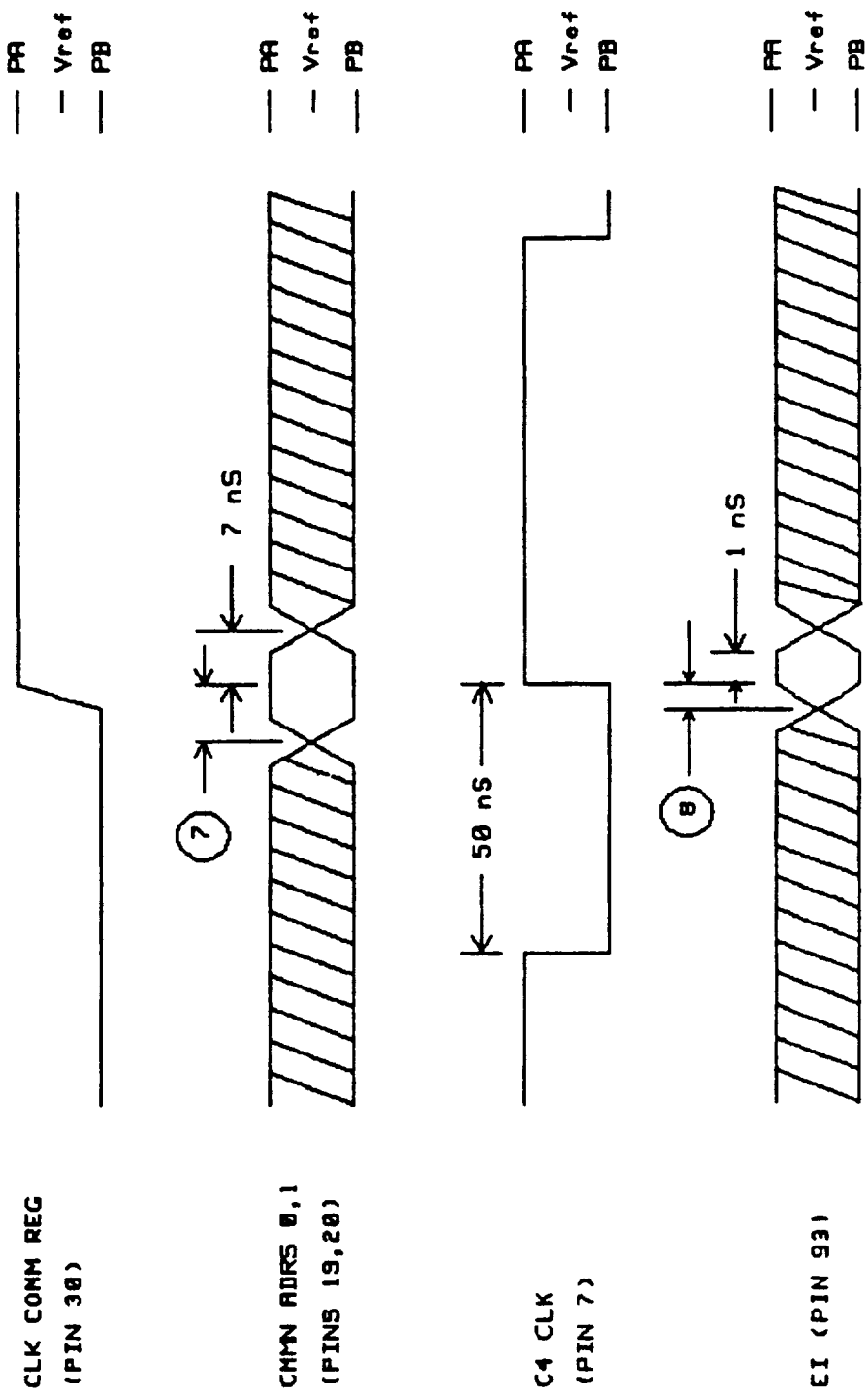


FIGURE 11b. Input timing parameters for table XV.

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NOTES:

1. (1) IOC CLK to BSTRP ENBL, DESTN=CM, ENBL T1. Hold time: minimum 14 nS.

2. (2) IOC CLK to DECR (R1). Hold time: minimum 43 nS.

3. (3) IOC CLK to IOCA REQ. Hold time: minimum 38 nS.

4. (4) IOC CLK to CHAN 3, CHAN 2, CHAN 1, CHAN 0, IOC BUS 1, CH/INTR HIT. Hold time: minimum 46 nS.

5. (5) ENBL CM WR to IOC CLK. Setup time: minimum 15 nS.

6. (6) DEST CM ADDR to IOC CLK. Setup time: minimum 15 nS.

7. (7) CMMN ADRS 0, CMMN ADRS 1 to CLK COMM REG. Setup time: minimum 15 nS.

8. (8) EI to C4 CLK. Setup time: minimum 3 nS.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (425 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (425 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (425 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (425 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	2.46 A	NA	Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 Q1-Q4 Q1-Q4
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 100 mV ± 100 mV ± 100 mV	 Q1,Q3 Q2,Q4 Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

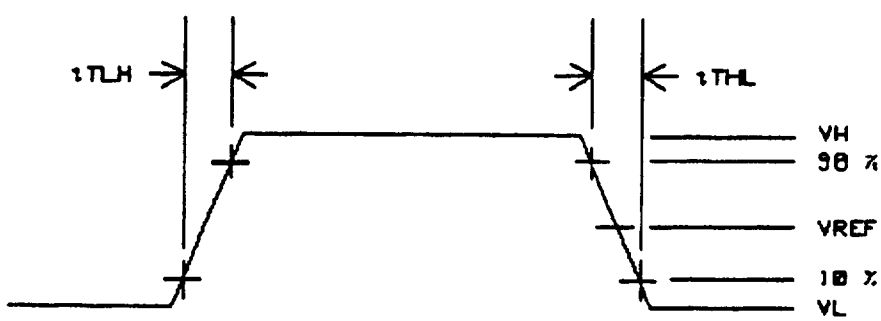
(b) The output compare levels shall be as follows:

High level: Vref ≤ Voh ≤ Vcc
Low level: 0.0 V ≤ Vol ≤ Vref

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	th	IOC CLK to BSTRP ENBL	14	14	14	-	-	-	nS	12
1	th	IOC CLK to DESTN=CM	14	14	14	-	-	-	nS	12
1	th	IOC CLK to ENBL T1	14	14	14	-	-	-	nS	12
2	th	IOC CLK to DECR (R1)	43	43	43	-	-	-	nS	12
3	th	IOC CLK to IOCA REQ	38	38	38	-	-	-	nS	12
4	th	IOC CLK to CHAN 3	46	46	46	-	-	-	nS	12
4	th	IOC CLK to CHAN 2	46	46	46	-	-	-	nS	12
4	th	IOC CLK to CHAN 1	46	46	46	-	-	-	nS	12
4	th	IOC CLK to CHAN 0	46	46	46	-	-	-	nS	12
4	th	IOC CLK to IOC BUS 1	46	46	46	-	-	-	nS	12
4	th	IOC CLK to CH/INTR HIT	46	46	46	-	-	-	nS	12
5	tsu	ENBL CM WR to IOC CLK	15	15	15	-	-	-	nS	12
6	tsu	DEST CM ADDR to IOC CLK	15	15	15	-	-	-	nS	12
7	tsu	CMMN ADRS 0 to CLK COMM REG	15	15	15	-	-	-	nS	12
7	tsu	CMMN ADRS 1 to CLK COMM REG	15	15	15	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8	tsu	EI to C4 CLK	3	3	3	-	-	-	nS	12
9	tpd	MAE ENBL to ADRS 7	-	-	-	18	18	18	nS	12
9	tpd	MAE ENBL to ADRS 6	-	-	-	18	18	18	nS	12
9	tpd	MAE ENBL to ADRS 5	-	-	-	18	18	18	nS	12
9	tpd	MAE ENBL to ADRS 4	-	-	-	18	18	18	nS	12
9	tpd	MAE ENBL to ADRS 3	-	-	-	18	18	18	nS	12
9	tpd	MAE ENBL to ADRS 2	-	-	-	18	18	18	nS	12
9	tpd	MAE ENBL to ADRS 1	-	-	-	18	18	18	nS	12
9	tpd	MAE ENBL to ADRS 0	-	-	-	18	18	18	nS	12
10	tpd	ENBL 10-15 to ADRS/WR 15	-	-	-	18	18	18	nS	12
10	tpd	ENBL 10-15 to ADRS/WR 14	-	-	-	18	18	18	nS	12
10	tpd	ENBL 10-15 to ADRS/WR 13	-	-	-	18	18	18	nS	12
10	tpd	ENBL 10-15 to ADRS/WR 12	-	-	-	18	18	18	nS	12
10	tpd	ENBL 10-15 to ADRS/WR 11	-	-	-	18	18	18	nS	12
10	tpd	ENBL 10-15 to ADRS/WR 10	-	-	-	18	18	18	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
11	tpd	ENBL CM/MAR to MAR 5	-	-	-	32	32	32	nS	12
11	tpd	ENBL CM/MAR to MAR 6	-	-	-	32	32	32	nS	12
11	tpd	ENBL CM/MAR to MAR 7	-	-	-	32	32	32	nS	12
11	tpd	ENBL CM/MAR to MAR 2	-	-	-	32	32	32	nS	12
11	tpd	ENBL CM/MAR to MAR 4	-	-	-	32	32	32	nS	12
11	tpd	ENBL CM/MAR to MAR 3	-	-	-	32	32	32	nS	12
11	tpd	ENBL CM/MAR to MAR 1	-	-	-	32	32	32	nS	12
11	tpd	ENBL CM/MAR to MAR 0	-	-	-	32	32	32	nS	12
12	tpd	ENBL ADR 0-9 to ADRS WR 9	-	-	-	18	18	18	nS	12
12	tpd	ENBL ADR 0-9 to ADRS WR 8	-	-	-	18	18	18	nS	12
12	tpd	ENBL ADR 0-9 to ADRS WR 2	-	-	-	18	18	18	nS	12
12	tpd	ENBL ADR 0-9 to ADRS WR 3	-	-	-	18	18	18	nS	12
12	tpd	ENBL ADR 0-9 to ADRS WR 4	-	-	-	18	18	18	nS	12
12	tpd	ENBL ADR 0-9 to ADRS WR 5	-	-	-	18	18	18	nS	12
12	tpd	ENBL ADR 0-9 to ADRS WR 6	-	-	-	18	18	18	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
12	tpd	ENBL ADR 0-9 to ADRS WR 7	-	-	-	18	18	18	nS	12
13	tpd	IOC CLK to BYTE SEL BIT	-	-	-	67	67	67	nS	12

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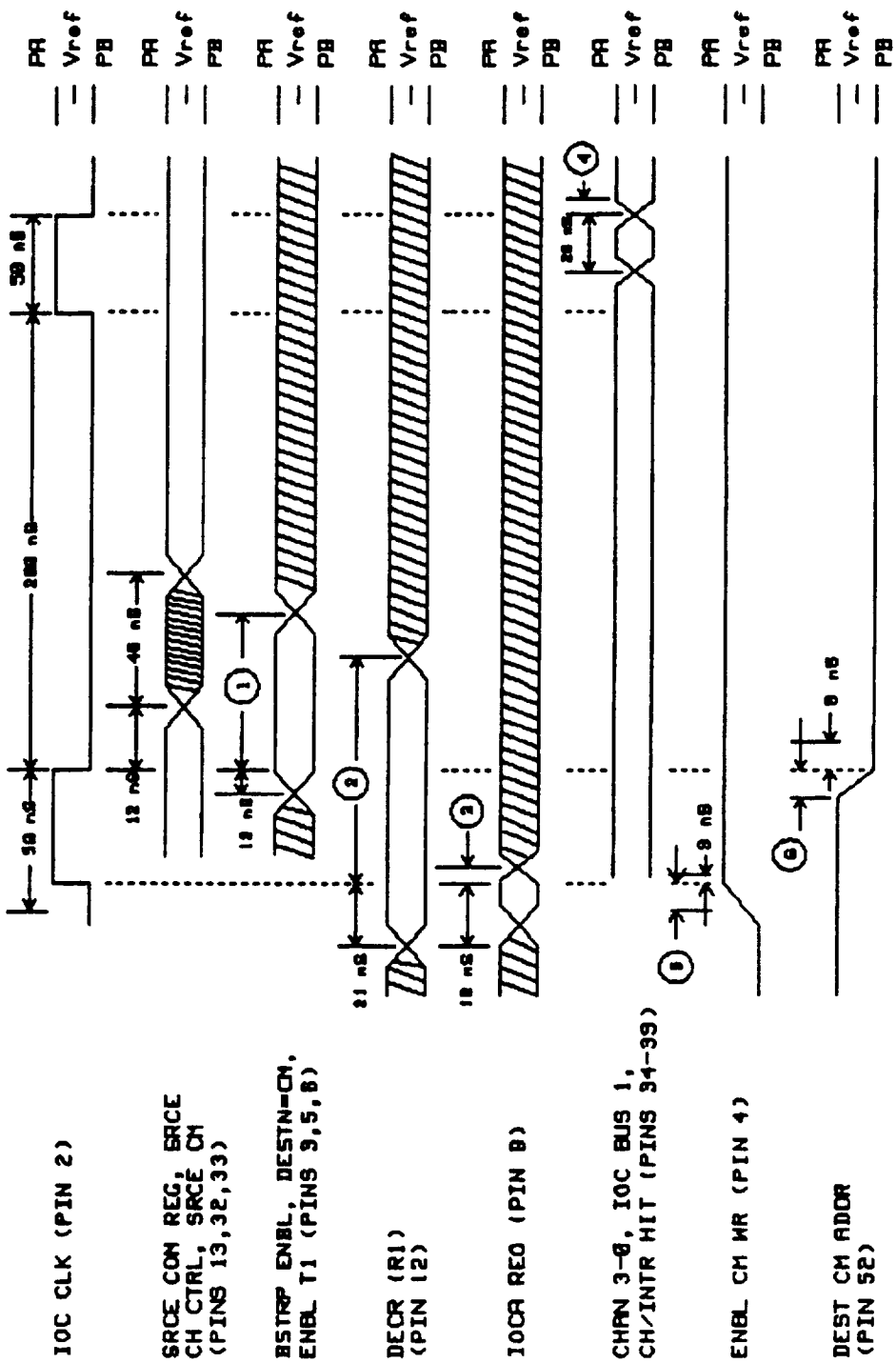


FIGURE 12a. Dynamic waveforms for table XVII.

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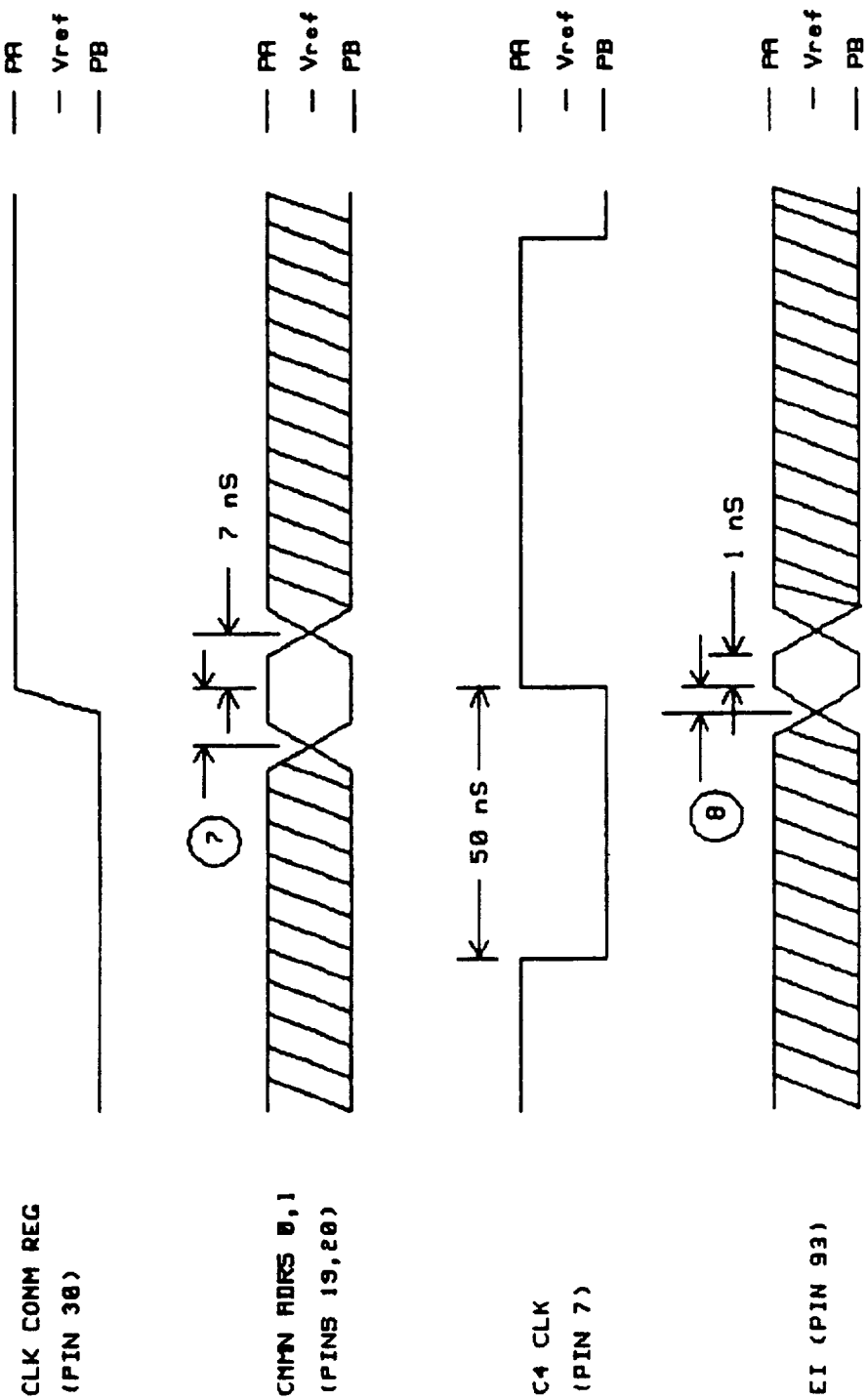


FIGURE 12b. Dynamic waveforms for table XVII.

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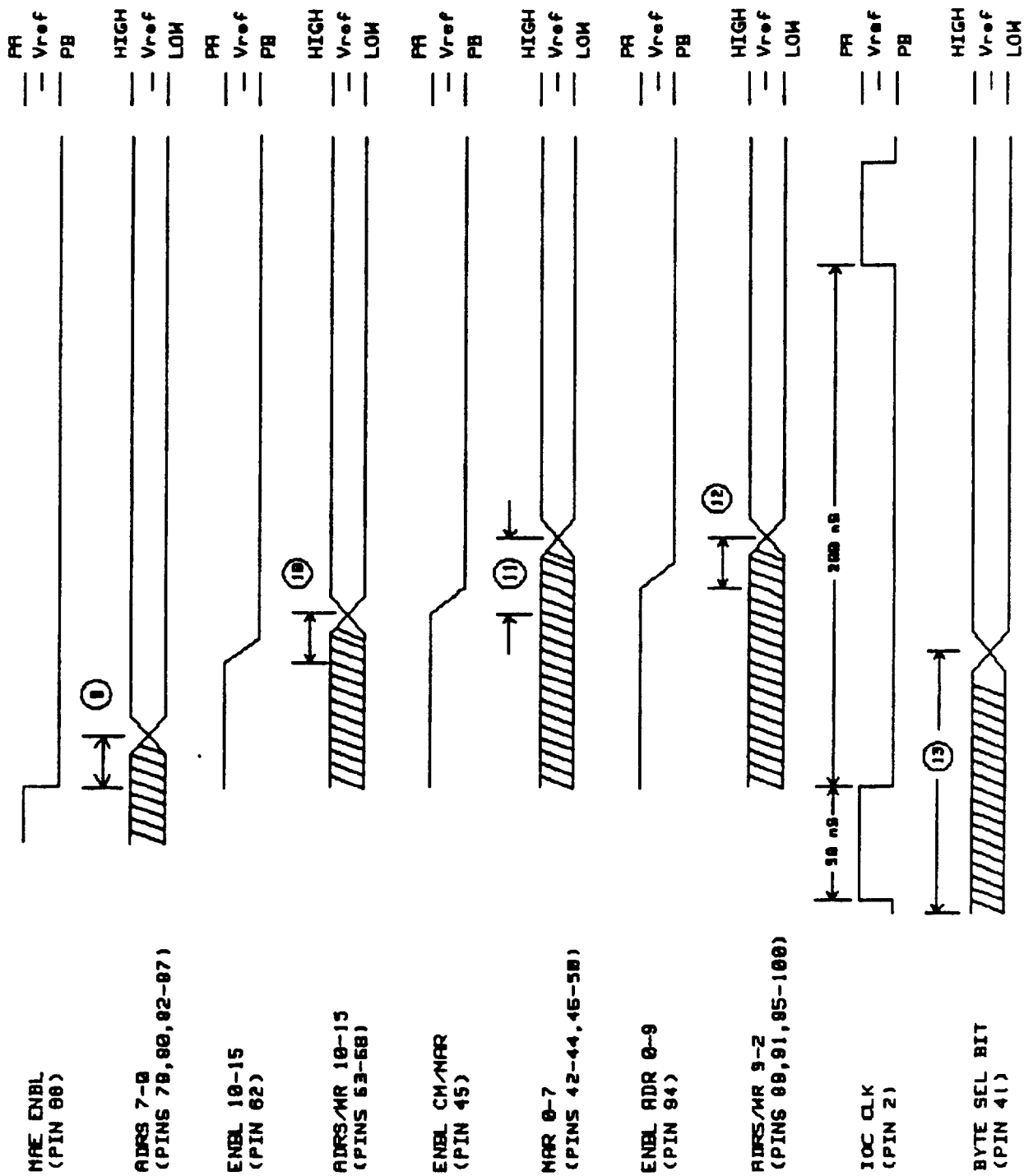


FIGURE 12c. Dynamic waveforms for table XVII.

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TABLE XVIII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	IOC CLK	2	8	125	pF	1-4
R2	DESTN=CM	3	4	125	pF	1-4
R3	ENBL CM WR	4	2	100	pF	1-4
R4	BSTRP ENBL	5	4	75	pF	1-4
R5	ENBL T1	6	4	75	pF	1-4
R6	C4 CLK	7	2	75	pF	1-4
R7	BYTE MODE	8				
R8	IOCA REQ	9	2	75	pF	1-4
R9	NDRO BIT MDE	11	2	100	pF	1-4
R10	DECR (R1)	12	2	75	pF	1-4
R11	SRCE COM REG	13	4	100	pF	1-4
R12	T4	14	2	75	pF	1-4
R13	IOC SEL 0	16	4	125	pF	1-4
R14	IOC SEL 1	17	4	125	pF	1-4
R15	CMMN ADRS 1	19	2	75	pF	1-4
R16	CMMN ADRS 0	20	2	75	pF	1-4
R17	CLK COMM REG	30	4	100	pF	1-4
R18	SRCE CH CTRL	32	4	125	pF	1-4
R19	SRCE CM	33	4	100	pF	1-4
R20	CHAN 3	34	6	100	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	CHAN 2	35	6	100	pF	1-4
R22	CHAN 1	36				
R23	CHAN 0	37	6	100	pF	1-4
R24	IOC BUS 1	38	2	75	pF	1-4
R25	CH/INTR HIT	39	6	100	pF	1-4
R26	ENBL CM/MAR	45	4	100	pF	1-4
R27	DEST CM ADDR	52	2	125	pF	1-4
R28	ENBL 10-15	62	4	75	pF	1-4
R29	MAE ENBL	88				
R30	FRCE P 00	92	4	75	pF	1-4
R31	EI	93	6	100	pF	1-4
R32	ENBL ADR 0-9	94	4	75	pF	1-4
Bidirectionals						
R33	SCE/DEST 0	21	10	125	pF	1-4
R34	SCE/DEST 4	22	10	150	pF	1-4
R35	SCE/DEST 5	23	10	150	pF	1-4
R36	SCE/DEST 1	24	10	125	pF	1-4
R37	SCE/DEST 3	26	10	125	pF	1-4
R38	SCE/DEST 6	27	10	150	pF	1-4
R39	SCE/DEST 7	28	10	150	pF	1-4
R40	SCE/DEST 2	29	10	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
R41	SCE/DEST 15	53	8	100	pF	1-4
R42	SCE/DEST 14	54	8	125	pF	1-4
R43	SCE/DEST 13	55	8	125	pF	1-4
R44	SCE/DEST 12	56	8	100	pF	1-4
R45	SCE/DEST 11	57	8	125	pF	1-4
R46	SCE/DEST 10	58				
R47	SCE/DEST 9	59				
R48	SCE/DEST 8	61	8	125	pF	1-4
R49	ADRS/WR 15	63	4	75	pF	1-4
R50	ADRS/WR 14	64	4	100	pF	1-4
R51	ADRS/WR 13	65	4	75	pF	1-4
R52	ADRS/WR 12	66				
R53	ADRS/WR 11	67				
R54	ADRS/WR 10	68	4	75	pF	1-4
R55	ADRS/WR 9	89	4	125	pF	1-4
R56	ADRS/WR 8	91	4	150	pF	1-4
R57	ADRS/WR 2	95	4	100	pF	1-4
R58	ADRS/WR 3	96	4	100	pF	1-4
R59	ADRS/WR 4	97	4	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R60	ADRS/WR 5	98	4	75	pF	1-4
R61	ADRS/WR 6	99	2	75	pF	1-4
R62	ADRS/WR 7	100	2	75	pF	1-4

NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R62
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R62
Capacitance measurement	NA	± 5.85 pF	R1
	NA	± 6.05 pF	R2
	NA	± 4.9 pF	R3
	NA	± 3.55 pF	R4,R5
	NA	± 3.65 pF	R6-R8
	NA	± 4.9 pF	R9
	NA	± 3.65 pF	R10
	NA	± 4.8 pF	R11
	NA	± 3.65 pF	R12
	NA	± 6.05 pF	R13,R14
	NA	± 3.65 pF	R15,R16
	NA	± 4.8 pF	R17
	NA	± 6.05 pF	R18
	NA	± 4.8 pF	R19
	NA	± 4.7 pF	R20-R23
	NA	± 3.65 pF	R24

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Capacitance measurement	NA	± 4.7 pF	R25
	NA	± 4.8 pF	R26
	NA	± 6.15 pF	R27
	NA	± 3.55 pF	R28-R30
	NA	± 4.7 pF	R31
	NA	± 3.55 pF	R32
	NA	± 5.75 pF	R33
	NA	± 7.0 pF	R34,R35
	NA	± 5.75 pF	R36,R37
	NA	± 7.0 pF	R38,R39
	NA	± 3.25 pF	R40
	NA	± 4.6 pF	R41
	NA	± 5.85 pF	R42,R43
	NA	± 4.6 pF	R44
	NA	± 5.85 pF	R45-R48
	NA	± 3.55 pF	R49
	NA	± 4.8 pF	R50
	NA	± 3.55 pF	R51-R54
	NA	± 6.05 pF	R55
	NA	± 7.3 pF	R56
	NA	± 4.8 pF	R57,R58
	NA	± 3.55 pF	R59,R60
	NA	± 3.65 pF	R61,R62

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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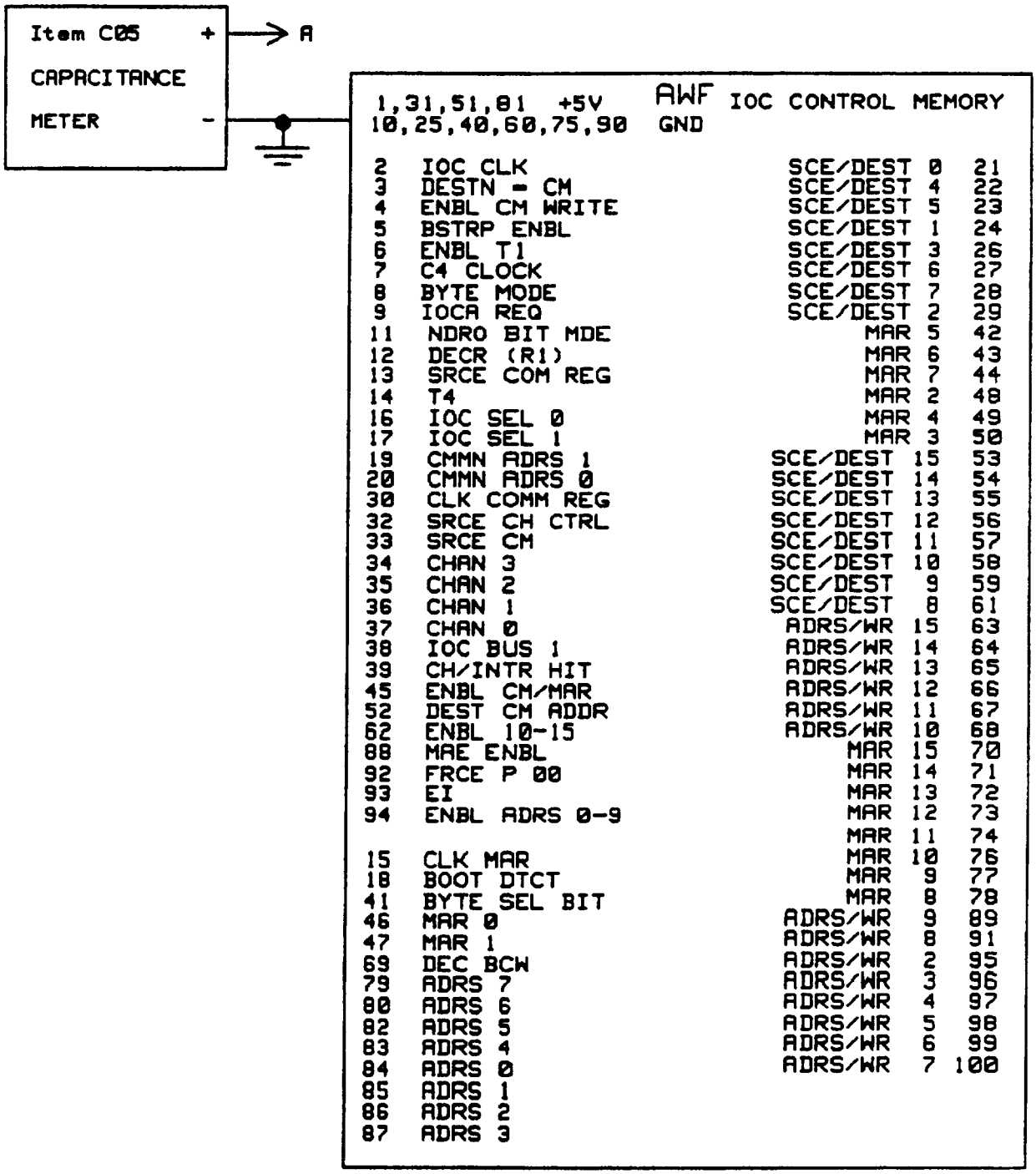


FIGURE 13. Input and output capacitance test circuit for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
T1	PSTA	1	0	400	mV	1-4	14,15
T2	PSTA	51	0	400	mV	1-4	14,15

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
Power supply current limit	2.46 A	NA	T1,T2
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1,T2 T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

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NOTES: - Continued.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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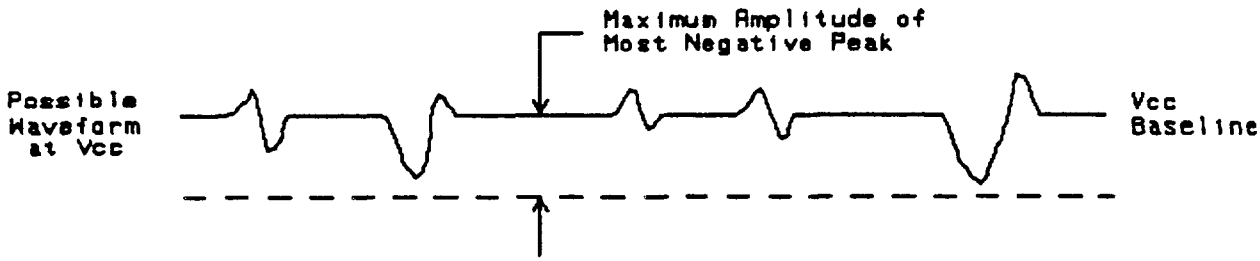
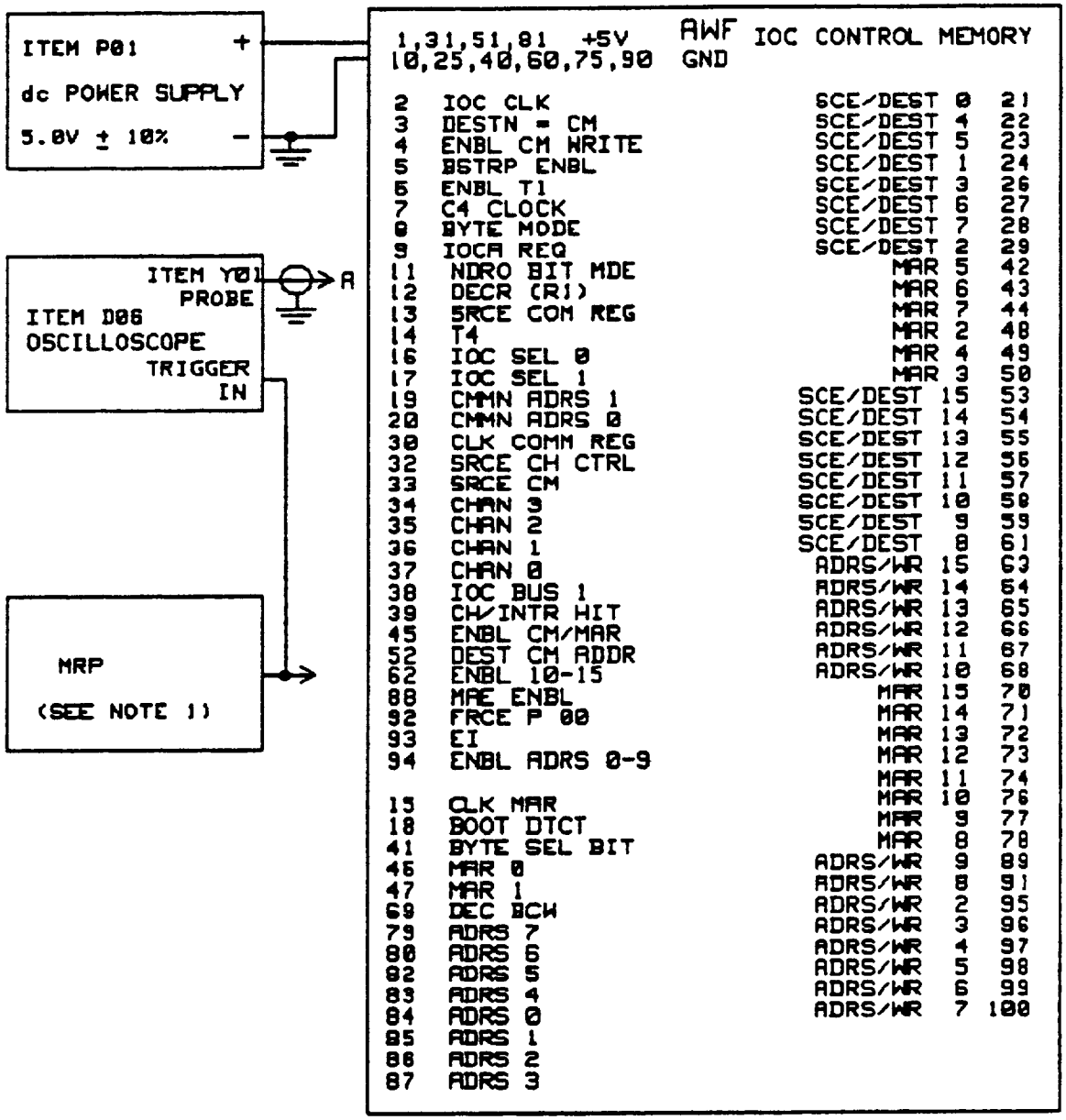


FIGURE 14. Power supply transient amplitude waveform for table XIX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XIX.

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